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FINAL THESIS

Progressive and degressive nonlinearities in heat conduction in
supercritical fluids

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1 Introduction

The liquid and gas states of matter are only defined within a certain region of the thermodynamic state space spanned by pressure and temperature. These two phases are separated by the binodal (also known as the coexistence curve) curve, which, however, terminates at a single, material-dependent point in the state space. This point is the liquid-vapor critical point of the substance. Above the pressure and temperature corresponding to the critical point, the material simultaneously exhibits liquid-like and gas-like properties; therefore, we cannot unambiguously refer to it as either a liquid or a gas. In this state, the medium is referred to as a supercritical fluid. This division of the thermodynamic state space is illustrated in Figure 1.

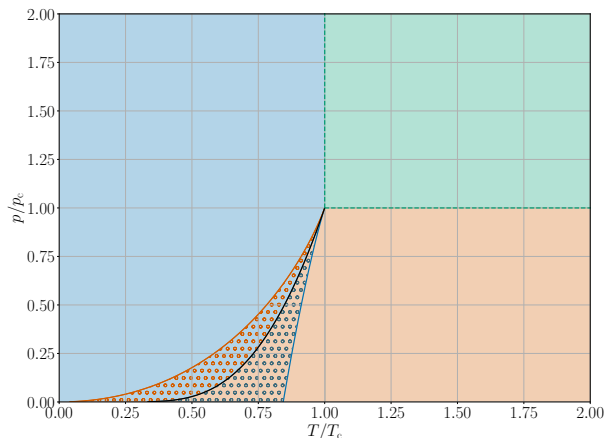


Figure 1: Classification of liquids (*blue*), gases (*red*) and supercritical fluids (*green*) as a function of pressure p/p_c and temperature T/T_c normalized by their critical values. The black solid line denotes the binodal curve. The diagram also shows the metastable phases and the spinodal curve.

Supercritical fluids are increasingly gaining prominence in the design of modern energy systems, serving both as working fluids and as coolants. One potential application of supercritical fluids is in Generation IV supercritical water nuclear reactors (SCWR) [1]. Their advantage over traditional reactors is that the efficiency of the Rankine cycle increases due to the higher operating temperatures. Furthermore, it becomes possible to direct the fluid from the reactor core straight to the turbine, thereby significantly simplifying the system. Another application of supercritical fluids relevant to the energy sector is in Enhanced Geothermal Systems (EGS), using supercritical carbon dioxide as

the working fluid [2]. The working principle of Enhanced Geothermal Systems is to extract heat from hot dry rock by rendering the rock permeable through fracturing, and subsequently introducing the working fluid via an injection well. The fluid is heated by the rock and then reaches the surface through production wells, where the thermal energy of the working fluid can be utilized. The use of supercritical carbon dioxide in Enhanced Geothermal Systems has numerous advantages. Beyond the fact that geothermal energy is renewable, carbon dioxide captured from the atmosphere can also be utilized by this technology. Carbon dioxide undergoes a smaller pressure drop when extracted from the rock than water, thus requiring less energy to operate the system. Its lower viscosity allows for a higher mass flow rate compared to water, and the resulting strong natural circulation also reduces the energy requirements of the system. Carbon dioxide does not react significantly with rock, thus reducing corrosion on the walls of the wells and surface equipment. By coupling Enhanced Geothermal Systems with the Organic Rankine Cycle (ORC), even electricity generation becomes possible.

In practice, however, the vicinity of the fluid's critical point is avoided, because fluids become highly compressible in this region, and their thermophysical properties exceed their liquid or gas state values by orders of magnitude while also exhibiting significant state dependence (see Figure 2). Consequently, even relatively small temperature changes may cause significant thermal expansion or substantial pressure increases, which complicates safe design. Since the interpretation of liquid and gas states ceases to exist, the phenomenon of boiling crisis, for instance, does not occur in supercritical fluids. At the same time, the sudden changes in material parameters can lead to the deterioration of the heat transfer coefficient.

Figure 2 also highlights that quantities diverging at the critical point (e.g., isobaric specific heat, isothermal compressibility) exhibit local extrema at different pressure-temperature pairs as one moves away from the critical point. Therefore, the anomalous behavior of material properties is not limited solely to the critical point but persists farther away as well. The pressure-temperature pairs characterizing the local extrema of these material properties define the so-called Widom lines, while the area enclosed by the Widom lines is the so-called Widom region, also known as the pseudocritical region. The Widom lines converge at the critical point, and can therefore be regarded as extensions of the binodal curve into the supercritical region, as illustrated in Figure 3. Sudden increases

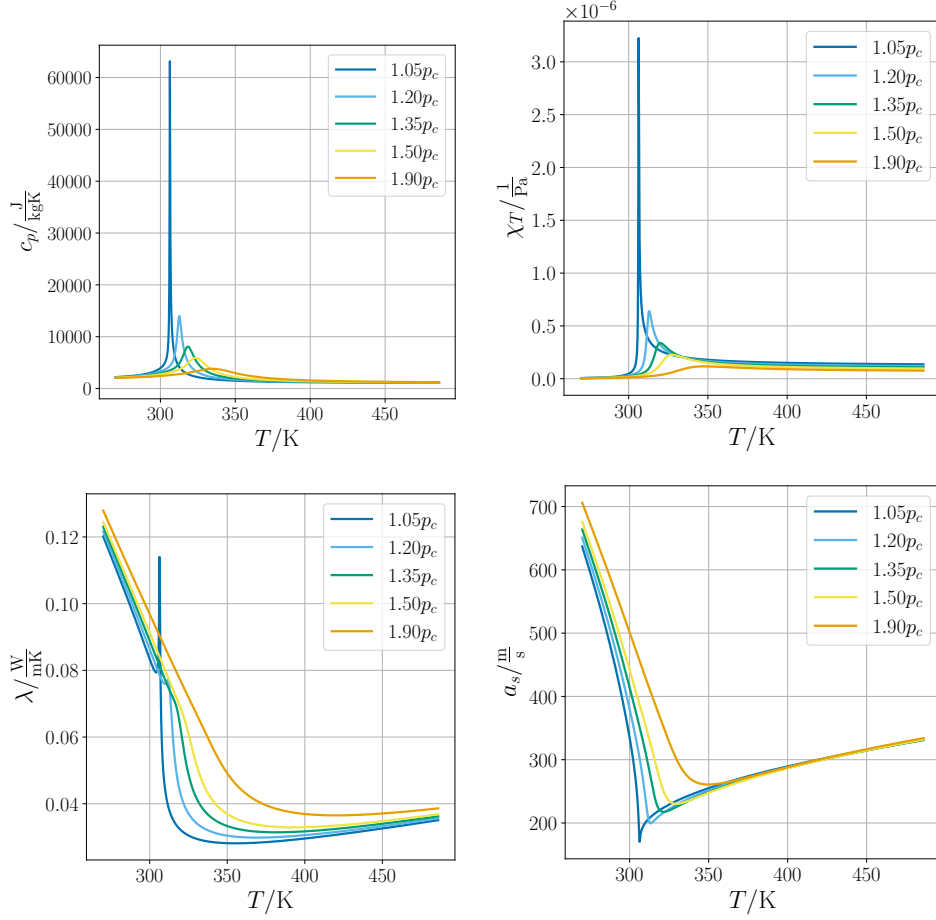


Figure 2: Some thermophysical properties of carbon dioxide in the vicinity of the critical point. The critical point of carbon dioxide is: $p_c = 7.3773$ MPa, $T_c = 304.1282$ K. *Top row*: isobaric specific heat and isothermal compressibility. *Bottom row*: thermal conductivity and isentropic speed of sound as a function of temperature at various supercritical pressures, based on the Span–Wagner equation of state.

in thermophysical properties are observed during phase transitions in subcritical fluids. Thus, by analogy, the Widom lines can be viewed as if a phase transition were occurring within the supercritical fluid at these states [3]. Every fluid possesses this property. Moreover, by applying appropriate nondimensionalization, these curves nearly coincide for different materials. This offers the opportunity to formulate scaling laws, which allow us to infer the properties of one fluid in a given state based on the properties of another fluid in the same state (see Figure 3).

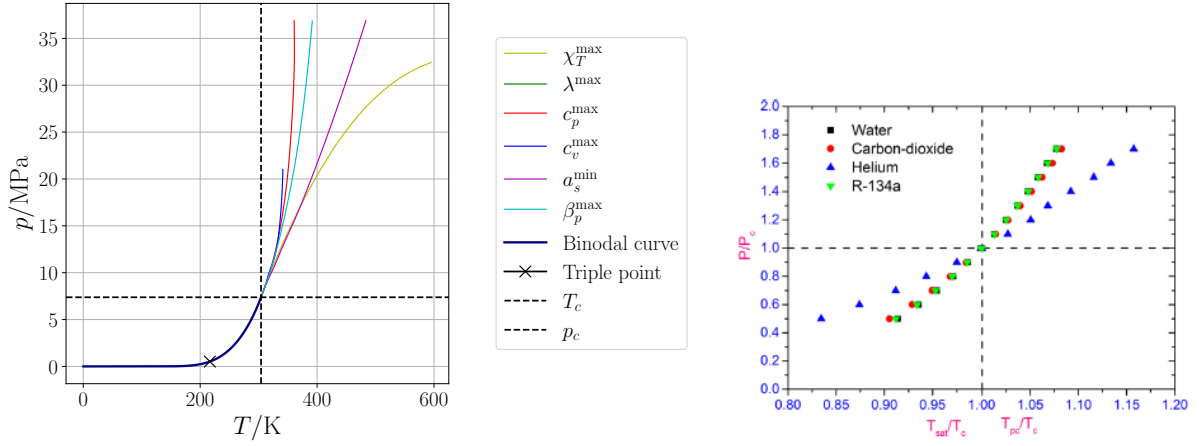


Figure 3: *Left*: Widom lines as extensions of the binodal curve calculated from the Span–Wagner equation of state. *Right*: Widom lines in the dimensionless state space [3].

[4] highlight the possible practical aspects of Widom anomalies. Their main focus is on adiabatic expansion and compression, which frequently occur in power cycles. The main conclusion of the work is that the knowledge of Widom lines can be used to design processes that behave more robustly against perturbations by selecting appropriate adiabats. Conversely, if an incorrect path is chosen, the process may become unstable.

In the vicinity of the critical point, the high compressibility of the fluid and the rapid variation of thermophysical properties cause a strong coupling between thermal and mechanical processes. This coupling has a profound influence on heat-transfer behavior. One manifestation of this is the piston effect. As a thermal boundary layer forms near a heated surface, it expands due to the large isobaric thermal expansion coefficient. This expansion compresses the more distant regions of the fluid in a piston-like manner. The resulting nearly isentropic pressure increase leads to an almost instantaneous rise in temperature, in contrast to the much slower diffusive propagation of heat. This phenomenon is expected to be present in the Widom region as well.

In this thesis, we develop and test a numerical method for investigating heat conduction in the Widom region. This method enables the investigation of the effect of material nonlinearities on heat conduction.

2 The piston effect

Classical heat conduction phenomena can be characterized by a single time scale. Using the ℓ_{char} characteristic length, λ thermal conductivity, c_p isobaric specific heat capacity and ρ density

$$\tau_{\text{diff}} = \frac{\ell_{\text{char}}^2}{\lambda/\rho c_p} \quad (1)$$

defines the diffusion time scale. Although both the thermal conductivity and the isobaric heat capacity diverge in the vicinity of the critical point, the $\frac{\lambda}{\rho c_p}$ thermal diffusivity tends to zero, because the specific heat tends to infinity faster than the thermal conductivity. Based on this, a strong slowing down of heat conduction phenomena is expected in the vicinity of the critical point. However, experiments performed on Earth observed rapid temperature equilibration in supercritical fluids, which contradicts the theoretical prediction. The contradiction was explained by buoyancy-driven convective mixing. However, during a space experiment in 1985, rapid temperature homogenization similar to measurements performed on Earth was observed even in the absence of gravity.

The phenomenon was explained analytically by Onuki and coworkers [5]: in a closed container of constant volume, the fluid near the heated surface expands and isentropically compresses the rest of the fluid, which warms up as a result. This process produces a much faster temperature equilibration than what would follow from heat conduction alone. They modeled the phenomenon such that, under the effect of the introduced heat quantity ΔQ , the temperature rise in a small V_1 subvolume is $\Delta T_1 = \frac{\Delta Q}{V_1 c_p}$, which causes the subvolume to expand by $\Delta V_1 = \beta_p V_1 \Delta T_1$. Since the fluid is located in a closed space, the total volume is constant; therefore, the pressure rises adiabatically, which ultimately results in a temperature rise in the remaining volume. The expansion of the V_1 subvolume is significant near the critical point due to the large value of the isobaric thermal expansion coefficient β_p and thus the temperature rise is also substantial.

In another experiment also performed under microgravity conditions, Straub and coworkers [6] investigated the heating of the surface of a sulfur hexafluoride fluid near its critical point. They showed that the temperature indeed homogenized extremely rapidly in the fluid, confirming Onuki's theory of critical speeding up, which has since been referred to in the literature as the piston effect.

During the experiment, the sulfur hexafluoride medium was contained within a spherical copper HPT-HYDRA type calorimeter with an internal radius of $R_s = 9.6$ mm, and a wall thickness of 0.4 mm (see in Figure 4). The spherical symmetry resulting from the spherical shape allowed for temperature measurements to be performed solely in the radial direction. Copper possesses excellent thermal conductivity; therefore, it was sufficient to heat the calorimeter wall using a heating wire along a single diameter during the heat pulse, which resulted in an approximately uniform wall temperature. Measurements confirmed that the wall temperature was indeed nearly uniform. The maximum temperature difference measured in the wall was 1 mK, observed at the end of the heat pulse. During the experiment, temperature was measured using four thermistors. One was placed on the wall of the calorimeter, while three were located inside the calorimeter within the fluid, at distances of 3.2 mm, 6.0 mm and 8.4 mm from the wall. The experiment was conducted under microgravity conditions in space. Consequently, a high vacuum could easily be established around the calorimeter to prevent convective heat losses, while the calorimeter was coated with gold to prevent radiative heat losses. During the experiment, the heating wire was glued to the calorimeter wall. Due to the adhesive's thermal conductivity, this setup introduced a dead time into the system. This means the calorimeter wall began to heat up with a delay relative to the start of the heating, and the temperature continued to rise for a while even after the heating had ended.

The results of the experiment were compared with a numerical simulation, in which the equations derived by Boukari and coworkers were solved using the initial and boundary conditions applied in the experiment [7]. The results of the numerical simulation closely approximated the measurement results. In the Boukari equations, pressure is independent of the spatial coordinate; that is, it does not account for the fact that the pressure change caused by the piston effect propagates in the medium as a wave at the speed of sound. This is justified by the fact that although the isentropic speed of sound a_s tends to 0 at the critical point, even at a very small $(T - T_c)/T_c > 10^{-9}$ distance from the critical point, the speed of sound is already 10 m/s. This results in an acoustic time scale that is sufficiently small compared to the diffusion time scale (and the duration of the heating).

$$\tau_a = \frac{\ell_{\text{char}}}{a_s} \quad (2)$$

Thus, considering the characteristic length scale of the experiment, wave phenomena conclude before the heating induces further change. The Straub experiment also validated

this simplification. During the experiment, thermistors were placed at several different points in the fluid. It was observed that the extent of heating was nearly identical at every thermistor. This indicates that the temperature increase in the fluid is nearly uniform, in agreement with Boukari's assumption that the associated pressure change is also approximately homogeneous and thus spatially independent. The results of the experimental and numerical calculations are illustrated in Figure 4.

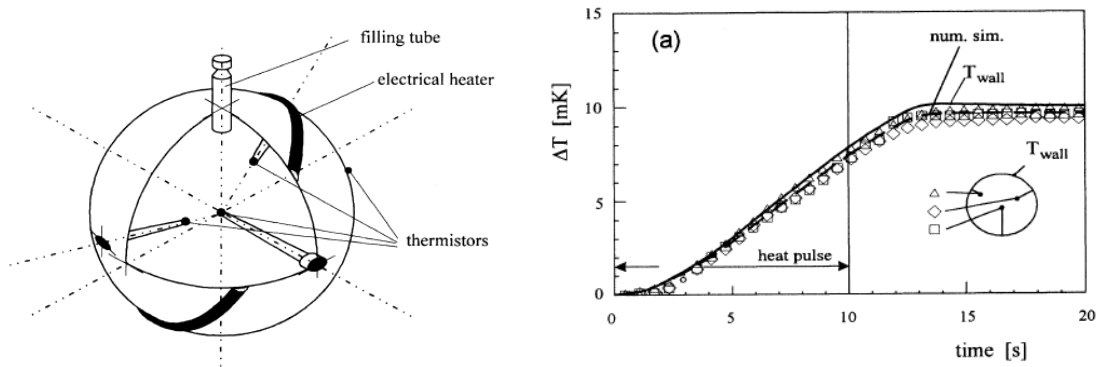


Figure 4: *Left*: schematic diagram of the HPT-HYDRA calorimeter; fluid temperature was recorded at three spatial points along the radius. *Right*: time evolution of the recorded temperatures for the initial temperature of $T_c + 0.1$ K [6].

3 Mathematical model of the piston effect

We base the numerical simulations constituting the main part of this thesis on the experiment presented in the previous chapter. However, since our aim is to investigate the material behavior in the simplest possible manner, we introduce the following simplifications. We employ a Cartesian coordinate system instead of spherical coordinates. To this end, instead of the spherical geometry, we investigate a cylindrical geometry with a length of $L = 100$ mm. We choose the radius of the cylinder such that the volume of the cylinder equals the volume of the calorimeter.

$$V_s = \frac{4\pi R_s^3}{3} = 3706 \text{ mm}^3 \quad (3)$$

$$V_s = V_c = R_c^2 \pi L \quad (4)$$

Here V_s is the volume of the sphere, V_c the volume of the cylinder, and R_c is the radius of the cylinder. From these, the radius of the cylinder can be determined as

$$R_c = \sqrt{\frac{V_s}{\pi L}} = 3.43 \text{ mm}. \quad (5)$$

Since the diameter of the cylinder defined in this way is negligible compared to the cylinder's length, we perform calculations only along the axial x coordinate axis of the cylinder during the numerical simulation. That is, we do not take into account the processes occurring along the radius of the cylinder, therefore, we approximate the problem in 1D. The heat input is applied on one of the cylinder's bases, and the heating time delay observed in the experiment is not modeled. The thermal insulation of the calorimeter is accounted for by prescribing a $\mathbf{\dot{q}} \cdot \mathbf{n} = 0$ normal heat flux density on every surface of the cylinder, except on the heated base during the period of heat input. The numerical implementation of this approach is detailed in the description of the boundary conditions of the governing equations.

3.1 Boukari approximation

Although the coupled thermomechanical processes of fluids are generally described by the simultaneous solution of the Navier–Stokes and heat conduction equations, Boukari was able to significantly reduce the governing equations using the following assumptions [7]:

- Due to microgravity, the body force density is zero.
- Since acoustic signal propagation is dominant only at the beginning of the process, the flow is neglected throughout the entire process; therefore, the material time derivative can be identified with the $\frac{\partial}{\partial t}$ partial time derivative .
- Based on the previous two conditions, the Navier–Stokes equation reduces to the

$$\nabla p = 0 \quad (6)$$

form. That is, the pressure is homogeneous throughout the entire process.

Based on these assumptions, the piston effect can be modeled by a heat conduction equation supplemented with a pressure correction term. The 1-dimensional spatial form of the equations is:

$$\rho c_p \frac{\partial T}{\partial t} = -\frac{\partial \dot{q}}{\partial x} + T \beta_p \frac{dp}{dt}, \quad (7)$$

$$\dot{q} = -\lambda \frac{\partial T}{\partial x}, \quad (8)$$

$$\frac{dp}{dt} = \frac{\int_0^L \rho \beta_p \frac{\partial T}{\partial t} dx}{\int_0^L \rho \chi_T dx}, \quad (9)$$

where T denotes the temperature of the fluid, β_p the isobaric thermal expansion coefficient, and χ_T the isothermal compressibility. Although the pressure is homogeneous, equation (9), derived from the mass balance, couples the pressure change to the temperature change. In highly compressible fluids, this coupling causes a significant pressure increase in response to a temperature increase, which feeds back into the heat conduction equation (7), thereby further increasing the temperature of the entire fluid.

3.2 Initial and boundary conditions

In the experiment, the fluid was initially in a homogeneous equilibrium state; therefore, we specify the initial p^0 pressure and initial T^0 temperature of the fluid at the $t = 0$ time as the initial condition. In accordance with the assumptions formulated earlier, we consider the investigated pipe section to be insulated, except during the period of heat input. In the experiment, surface heating was applied with constant power for a specified duration. In the case of the 1-dimensional geometry, the heat input is applied at the coordinate

$x = 0$ of the closed cylinder, while we prescribe an adiabatic boundary condition at the coordinate $x = L$:

$$\dot{q}(t, x = 0) = \begin{cases} \dot{q}_0, & \text{ha } 0 \leq t \leq t_h, \\ 0, & \text{ha } t_h \leq t, \end{cases} \quad \dot{q}(t, x = L) = 0, \quad (10)$$

where t_h is the duration of the heating phase and $\dot{q}_0$ is the heat flux density prescribed on the end face of the cylinder, which can be calculated from the heating power and the heated surface area based on the

$$\dot{q}_0 = \frac{\dot{Q}}{A} = \frac{\dot{Q}}{R_h^2 \pi} \quad (11)$$

formula. The amount of heat introduced during the heating phase is

$$Q = \dot{Q} t_h. \quad (12)$$

3.3 Thermophysical properties

Precise knowledge of thermophysical properties is required to solve the equations since, as we demonstrated earlier, thermophysical properties exhibit significant state dependence in the vicinity of the critical point. In this thesis the material we investigate is carbon dioxide, therefore, we selected the equation of state describing the material behavior accordingly. For describing the state dependence of thermophysical properties in the case of carbon dioxide, the NIST database uses the Span-Wagner [8] equation of state, which expresses the state dependence of various material properties using empirical correlations based on the Helmholtz specific free energy

$$f = e - Ts, \quad (13)$$

where e is the specific internal energy and s is the specific entropy. The specific free energy is a thermodynamic potential as a function of temperature and v specific volume; therefore, the specific entropy and the pressure can be calculated as

$$s(T, v) = - \left. \frac{\partial f}{\partial T} \right|_v, \quad p(T, v) = - \left. \frac{\partial f}{\partial v} \right|_T \quad (14)$$

partial derivatives. From these, the c_v isochoric specific heat capacity, the isobaric cubic thermal expansion coefficient, and the isentropic compressibility can be determined

directly:

$$c_v = T \left. \frac{\partial s}{\partial T} \right|_v, \quad \beta_p = -\frac{1}{v} \frac{\left. \frac{\partial p}{\partial T} \right|_v}{\left. \frac{\partial p}{\partial v} \right|_T}, \quad \chi_T = -\frac{1}{v} \frac{1}{\left. \frac{\partial p}{\partial v} \right|_T}, \quad (15)$$

while the isobaric specific heat capacity, the isentropic speed of sound, and the μ_{JT} Joule–Thomson coefficient can be determined from these due to thermodynamic consistency:

$$c_p = c_v + Tv \frac{\beta_p^2}{\chi_T}, \quad a_s = \sqrt{v \frac{c_p}{c_v} \frac{1}{\chi_T}}, \quad \mu_{JT} = \frac{v}{c_p} (T\beta_p - 1). \quad (16)$$

The Span–Wagner equation of state for carbon dioxide provides the specific free energy as a function of density and temperature. The following relationship exists between the partial derivatives with respect to specific volume and density:

$$\frac{\partial}{\partial v} = \frac{\partial}{\partial \rho} \frac{d\rho}{dv} = -\frac{1}{v^2} \frac{\partial}{\partial \rho} = -\rho^2 \frac{\partial}{\partial \rho} \quad (17)$$

Therefore, the above formulas can be readily rewritten in terms of the density. Since we used functions derived from the Span–Wagner equation of state for the simulation, we compared the implemented thermophysical properties with the values reported in the NIST database (see Figure 5). The agreement is satisfactory.

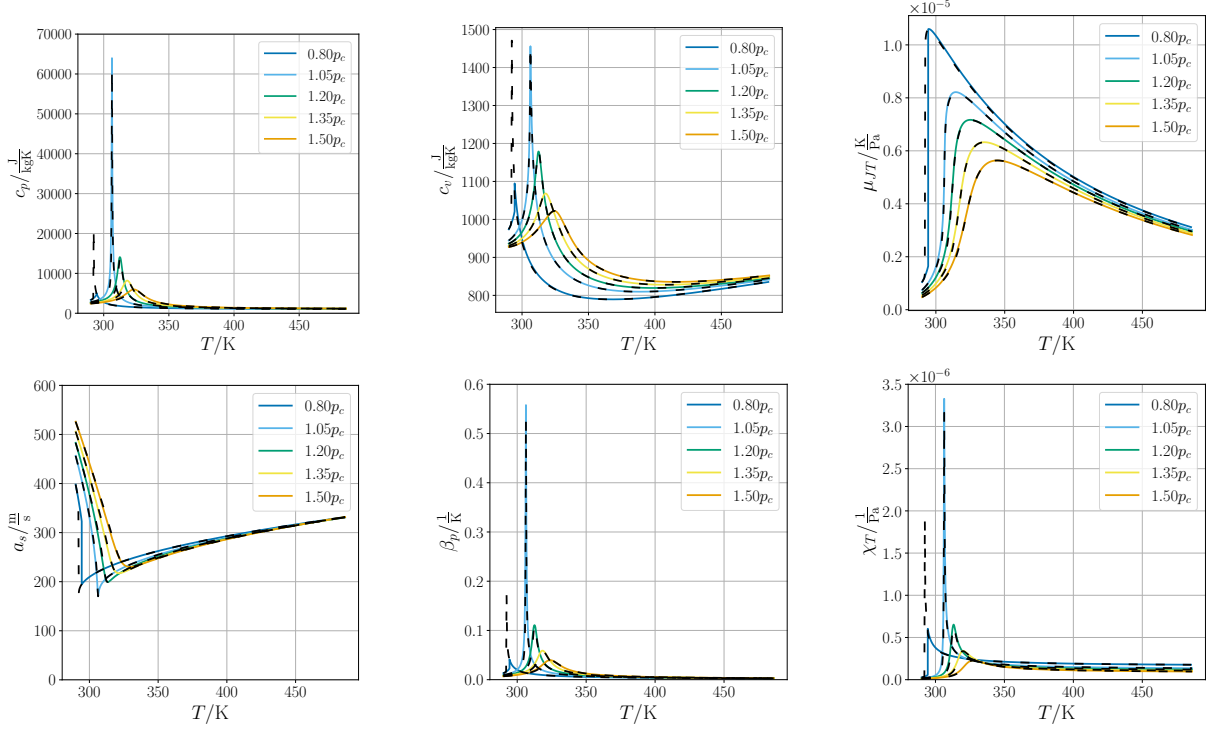


Figure 5: Thermophysical properties calculated from the Span-Wagner equation of state (black dashed line) and material properties reported in the NIST database (colored solid line) as a function of temperature along a few isobars. *Top row*: isobaric specific heat, isochoric specific heat, and Joule-Thomson coefficient. *Bottom row*: isentropic speed of sound, isobaric thermal expansion coefficient, and isothermal compressibility.

The thermal conductivity of carbon dioxide is also required for the simulation. However, this cannot be derived from a thermodynamic potential. At the same time, correlations approximating the thermal conductivity can be found in the literature. For the simulations, we used the correlation proposed by [9]

$$\lambda(\rho, T) = \lambda_0(T) + \Delta\lambda(\rho, T) + \Delta\lambda_c(\rho, T). \quad (18)$$

Here λ_0 is the thermal conductivity component corresponding to „0 density“, $\Delta\lambda_c(\rho, T)$ accounts for the divergence of the thermal conductivity in the vicinity of the critical point, $\Delta\lambda(\rho, T)$ represents all other effects. The terms λ_0 and $\Delta\lambda(\rho, T)$ are calculated solely from density and temperature at a given state according to

$$\lambda_0(T) = \frac{\sqrt{T/T_c}}{J \sum_{k=1} \frac{L_k}{(T/T_c)^k}}, \quad \Delta\lambda(\rho, T) = \sum_{i=1}^6 (B_{1,i} + B_{2,i}(T/T_c))(\rho/\rho_c)^i \quad (19)$$

formulas, where L_k , $B_{1,i}$ and $B_{2,i}$ are known constants. The determination of the critical component is significantly more complex; however, for engineering applications, at temperatures more than 10 K away from the critical point, the authors suggest the

$$\Delta\lambda_c(\rho, T) = \frac{-17.47 - 44.88\Delta T_c}{0.8563 - \exp[8.865\Delta T_c + 4.16\Delta\rho_c^2 + 2.302\Delta T_c\Delta\rho_c - \Delta\rho_c^3] - 0.4503\Delta\rho_c - 7.197\Delta T_c}, \quad (20)$$

empirical formula for determining the critical component of the thermal conductivity, where $\Delta T_c = (T/T_c) - 1$ and $\Delta\rho_c = (\rho/\rho_c) - 1$.

For the simulations, we calculated the critical component of the thermal conductivity based on the correlation (20). Disregarding the authors' suggestion, we applied this correlation over the entire temperature range, including the critical point; despite this, we obtained results within error margins acceptable for our purpose (see Figure 6).

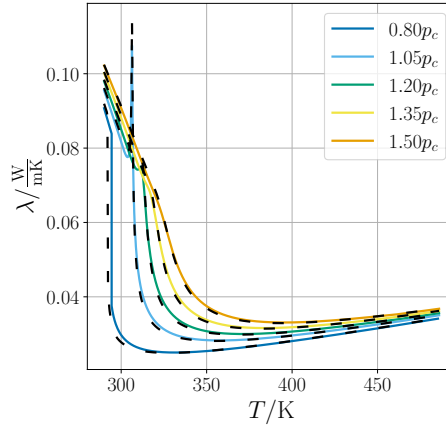


Figure 6: Thermal conductivity calculated based on [9] (black dashed line) and as reported in the NIST database (colored solid line) as a function of temperature along a few isobars.

4 Linear case

The Boukari equations are nonlinear differential equations; therefore, we linearize them. Apart from the state dependence of material properties, the only nonlinearity is the $T\beta_p \frac{dp}{dt}$ term appearing in the heat conduction equation (7). We linearize the equations by assuming that the thermophysical properties are constant and substitute their values with those of the initial state; furthermore, in the aforementioned nonlinear term, we replace the temperature with the initial temperature; thus, the governing equations simplify as follows:

$$\rho^0 c_p^0 \frac{\partial T}{\partial t} = -\frac{\partial \dot{q}}{\partial x} + T^0 \beta_p^0 \frac{dp}{dt}, \quad (21)$$

$$\frac{dp}{dt} = \frac{1}{L} \frac{\beta_p^0}{\chi_T^0} \int_0^L \frac{\partial T}{\partial t} dx, \quad (22)$$

$$\dot{q} = -\lambda^0 \frac{\partial T}{\partial x} \quad (23)$$

4.1 Dominant terms in the equations

Examining the above equations, we can identify dominant parameter combinations that significantly influence the dynamics of the process [10]. To derive these, we neglect spatial dependencies and rely solely on thermostatic relations. Since the sample is insulated during the experiment (apart from the heat input), the T^∞ temperature is homogeneous in the final state. Since the volume is constant, the First Law of Thermodynamics for an isochoric process

$$\rho_0 \frac{4\pi}{3} R_g^3 c_v^0 (T^\infty - T^0) = \int_0^\infty \dot{Q} dt = Q. \quad (24)$$

The volume appearing in the relationship corresponds to the calorimeter volume in the Straub experiment, which is identical to the cylinder volume employed in the simulation. By rearranging the above equation, we can identify which material properties have the greatest impact on the difference between the final and initial temperatures:

$$T^\infty - T^0 = \frac{Q}{\frac{4\pi}{3} R_g^3 \rho^0 c_v^0}. \quad (25)$$

The geometry and the heat input are specific to the experiment, not the material. Therefore, according to the linear estimation, the material parameter influencing the difference

between the final and initial temperatures is the reciprocal of the product of the isochoric specific heat and the density.

We can proceed similarly to determine the difference between the final and initial pressures. The process is isochoric and the initial and final temperatures are known, thus

$$p^\infty - p^0 = \frac{\beta_p^0}{\chi_T^0} (T^\infty - T^0) = \frac{Q}{\frac{4\pi}{3} R_g^3} \frac{\beta_p^0}{\chi_T^0 \rho^0 c_v^0}. \quad (26)$$

By expressing the isothermal compressibility in terms of the speed of sound, our previous result can be cast into the form

$$p^\infty - p^0 = \frac{Q}{\frac{4\pi}{3} R_g^3} \frac{\beta_p^0 (a_s^0)^2}{c_p^0} \quad (27)$$

where the appearing material parameter combination $\Gamma = \frac{\beta_p^0 (a_s^0)^2}{c_p^0}$ is precisely the dimensionless Grüneisen parameter.

Finally, we determine the isentropic temperature change occurring during the heating phase. We denote the state following the completion of the heating phase by the index 0^+ . Since the pressure change is only relevant during the heating phase, we consider the pressure to be constant after the completion of the heating phase; that is $p^\infty = p^{0+}$. Although heat transfer takes place during the heating phase, due to slow heat conduction, the fluid at the center of the spherical container and the fluid at the end of the cylinder opposite to the heat input in the numerical experiment is not heated; therefore, the process can be considered isentropic (cf. Onuki's explanation [5]). By transforming and linearizing the partial derivative $\left. \frac{\partial T}{\partial p} \right|_s$ the temperature change can be determined according to

$$T^{0+} - T^0 = \frac{Q}{\frac{4\pi}{3} R_g^3} \frac{T^0}{\rho^0} \left(\frac{\beta_p^0 a_s^0}{c_p^0} \right)^2 \quad (28)$$

formula. Figure 7 illustrates the state dependence of the identified dominant terms

$$\frac{T^0}{\rho^0} \left(\frac{\beta_p^0 a_s^0}{c_p^0} \right)^2 \quad \frac{\beta_p^0 (a_s^0)^2}{c_p^0} \quad \frac{1}{\rho^0 c_v^0} \quad (29)$$

in the vicinity of the critical point.

Note that in the vicinity of the critical point, the term $\frac{T^0}{\rho^0} \left(\frac{\beta_p^0 a_s^0}{c_p^0} \right)^2$ has an inflection point and exhibits a steep slope in this region. Consequently, even in the linear case, the temperature rise during the heat pulse for processes initiated from identical initial pressures but slightly different initial temperatures can differ significantly. It is also important to note that the parameter $\frac{\beta_p^0 (a_s^0)^2}{c_p^0}$ takes on the same value in different states.

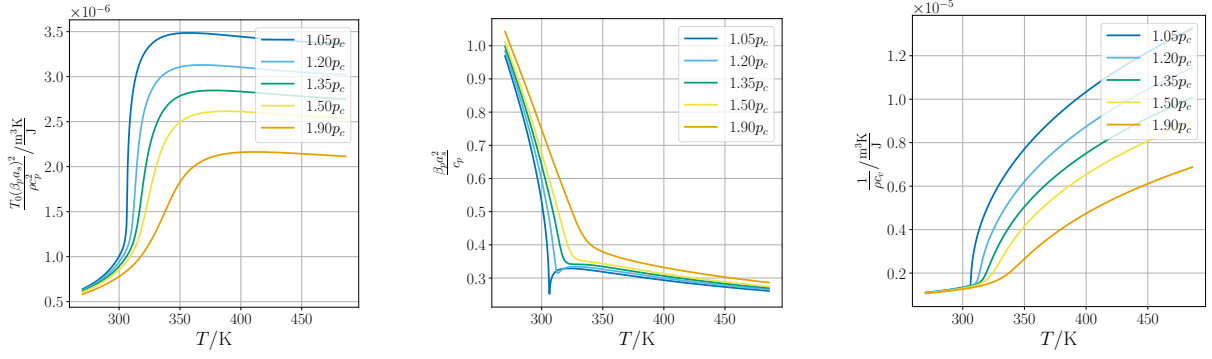


Figure 7: State dependence of material parameters influencing the piston effect in the vicinity of the critical point.

For example, the curves corresponding to pressures $1,05 p_c$ and $1,20 p_c$ intersect. This may result in a simulation initiated from the initial temperature T^0 corresponding to the intersection point and an initial pressure of $p^0 = 1,05 p_c$ behaving as if it had started from an initial pressure of $p^0 = 1,20 p_c$. Whether this is a numerical artifact resulting from the inversion of the equation of state or an actual physical behavior requires further investigation.

4.2 Discretization

We will solve equations (7), (8), and (9) describing the problem numerically. We employ equidistant discretization in both space and time, dividing the $x \in [0, L]$ domain into N cells of size $\Delta x = \frac{L}{N}$. We represent the spatially dependent quantities, namely temperature and heat flux density, using a spatial stagger of half a step relative to each other. Since heat flux density is prescribed at the boundaries, we assign integer spatial indices to the heat flux density and half-integer spatial indices to the temperatures. We implement the time integration using the explicit Euler method; thus, the discretized system of equations is:

$$T_{n+\frac{1}{2}}^{j+1} = T_{n+\frac{1}{2}}^j + \frac{1}{\rho^0 c_p^0} \frac{\Delta t}{\Delta x} (\dot{q}_n^j - \dot{q}_{n+1}^j) + \frac{T^0 \beta_p^0}{\rho^0 c_p^0} (p^{j+1} - p^j), \quad (30)$$

$$p^{j+1} = p^j + \frac{1}{L} \frac{\beta_p^0}{\chi_T^0} \sum_{n=0}^{N-1} (T_{n+\frac{1}{2}}^{j+1} - T_{n+\frac{1}{2}}^j) \Delta x, \quad (31)$$

$$\dot{q}_n^{j+1} = \lambda^0 \frac{T_{n-\frac{1}{2}}^{j+1} - T_{n+\frac{1}{2}}^{j+1}}{\Delta x}, \quad (32)$$

where j denotes the time index, n the spacial index, and Δt the time step. To maintain the stability of the explicit Euler method, the theoretical maximum of the permissible time step is $\Delta x^2 / \frac{2\lambda^0}{\rho^0 c_p^0} = \Delta x^2 / 2\alpha$, where α is the thermal diffusivity. Since this is a considerably large value in most cases, we applied a significantly finer temporal discretization to achieve a more accurate solution. To obtain a detailed resolution of the heat input duration, we prescribed that the duration of the heating phase be divided into a minimum of 10 time steps; that is, we determined the applied time step according to the

$$\Delta t = \min \left(\frac{\Delta x^2}{\frac{2\lambda^0}{\rho^0 c_p^0}}, \frac{t}{10} \right) \quad (33)$$

formula.

It can be observed that both (30) and (32) contain the temperature and pressure values at the new time instant, that is the terms $T_{n+\frac{1}{2}}^{j+1}$ and p_{j+1} thus, these two variables must be determined iteratively. During the iterative procedure, within a single time step, we first determine a temperature $T_{n+\frac{1}{2}}^{j+1}$ from equation (30) without the pressure correction term $\frac{T^0 \beta_p^0}{\rho^0 c_p^0} (p^{j+1} - p^j)$, and then, substituting this into equation (32), we determine an estimate for the pressure p^{j+1} . Subsequently, we substitute this just-determined pressure p^{j+1} into equation (30), which now includes the pressure correction term, to determine a new temperature $T_{n+\frac{1}{2}}^{j+1}$. This process continues until the $T_{n+\frac{1}{2}}^{j+1}$ temperature values and p^{j+1} determined in the two iteration steps become nearly identical. After this, $\dot{q}_n^{j+1}$ is calculated based on equation (31). In the next time step, the terms with index $j+1$ from the previous step become the terms with index j . Figure 8 presents a box diagram of the iterative process.

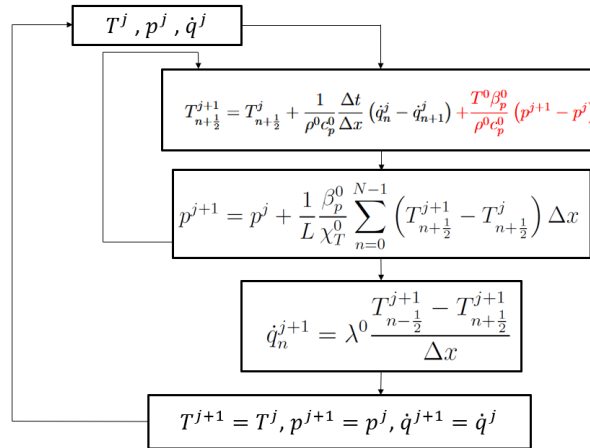


Figure 8: The iterative process

During the simulation, the geometry was divided into 20 elements of equal length along the x coordinate axis. This might appear insufficient at first glance; however, [11] presents a grid independence study for a similar problem and geometry, demonstrating that using 20 cells instead of 100 resulted in a deviation of less than 3% for all field variables. Increasing the number of cells would significantly increase the computational cost. We plan to repeat the calculations with a finer resolution on the Komondor supercomputer in the future.

4.3 Numerical investigations

In this section, we present the solution to the linear problem through a representative case. The initial conditions for the simulation are $p^0 = 7.38$ MPa and $T^0 = 305$ K; the heating power is $P = 3.85$ W and the heating duration is $t_h = 10$ s. Three time scales are present during the simulation. The acoustic time scale is $\tau_a = L/a_s$, over which wave phenomena take place. This essentially sets a lower limit on the duration of the heat input, as ideally, the heating time should be kept at least one or two orders of magnitude above this value. This is because the Boukari approximation is incapable of handling wave phenomena, and furthermore, the technological implementation of such a short heat pulse is questionable. The second time scale is the duration of the heat input, during which the piston effect dominates. The third time scale is the diffusion time scale $\tau_{\text{diff}} = L^2/\frac{\lambda}{\rho^0 c_p} = L^2/\alpha$ over which diffusion processes occur. We determined the diffusion time from the material properties defined by the initial conditions: $\tau_{\text{diff}} = L^2/\frac{\lambda^0}{\rho^0 c_p^0}$. The simulation ran until the end of the diffusion time scale to ensure that the temperature would be homogenized by the end of the simulation.

Figure 9 shows the time evolution of the pressure. It can be seen that in the linear case, the pressure increases linearly during the heating phase, and then settles at a constant value after the heating concludes. The pressure is homogeneous within the fluid, whereas the temperature is not. It is worth examining the cell furthest from the heat input, as the effect of direct heat conduction is minimal there; thus, the piston effect dominates most in this region. Figure 10 presents the time evolution of the temperature in the cell furthest from the location of the heat input. It is evident that in the linear case, the temperature increased linearly during the heat input, after which the diffusion process begins to dominate. The diagram demonstrates the accelerating effect of the piston effect

on temperature equilibration; a significant portion of the total temperature change occurs during the heating phase.

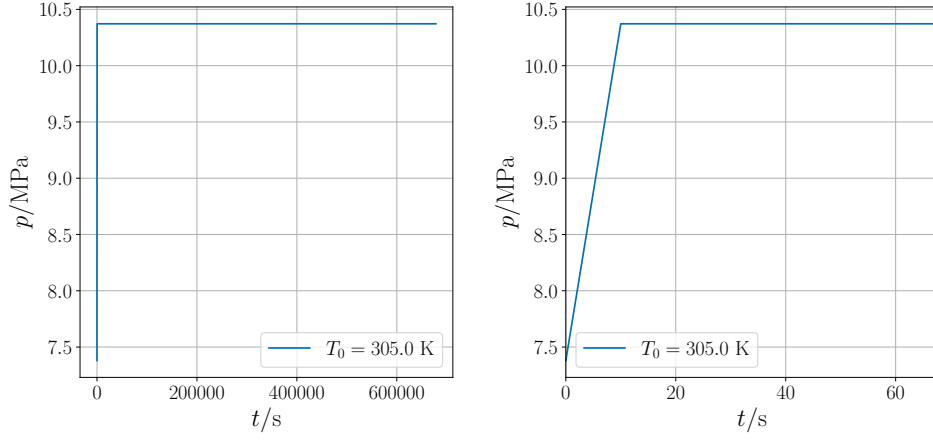


Figure 9: *Left*: Pressure change over the entire duration of the simulation. *Right*: Pressure change at the beginning of the process; the heating phase lasts 10 s.

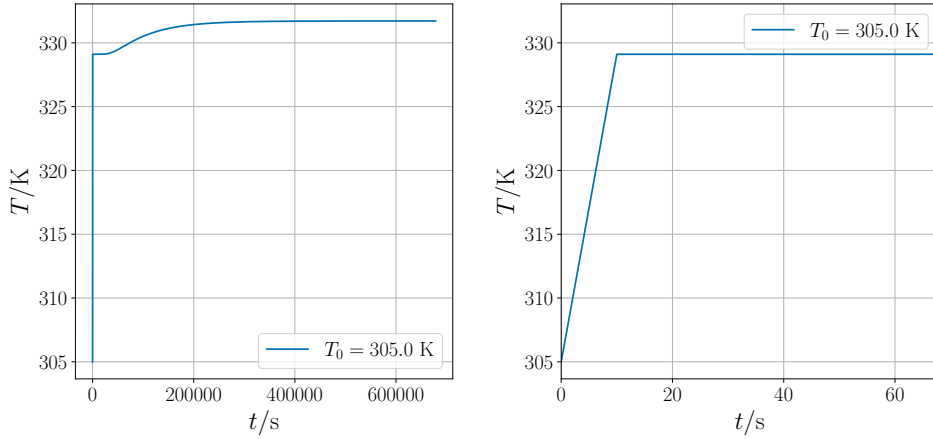


Figure 10: *Left*: Temperature change in the cell furthest from the heat input over the entire duration of the simulation. *Right*: Temperature change in the cell furthest from the heat input at the beginning of the process; the heating phase lasts 10 s.

Finally, we plot the temperature along the length of the cylinder at various time instances. This visualizes the process of temperature homogenization throughout the entire fluid. First, we depict the temperature as a function of the spatial coordinate during the heating phase and at time instances shortly after the heating phase. When simulating this representative case, we employed 100 cells to ensure smoother temperature distributions. It is evident that during the heating phase, the temperature increased not only in the directly heated cells but throughout the entire fluid; indeed, halfway through

the duration of the heat input, the temperature is nearly 320 K in every cell. This warming is caused by the piston effect, as only a fraction of the diffusion time has elapsed by this point. These are illustrated in Figure 11. Subsequently, we examined the diffusion phase; the temperature distributions during this phase are shown in Figure 12. When plotting the time evolution of the temperature, it was apparent that the temperature had already homogenized by half of the $\tau_{\text{diff}} = 678\,208\text{ s}$ diffusion time. We selected the specific time instances for plotting the temperature along the length accordingly.

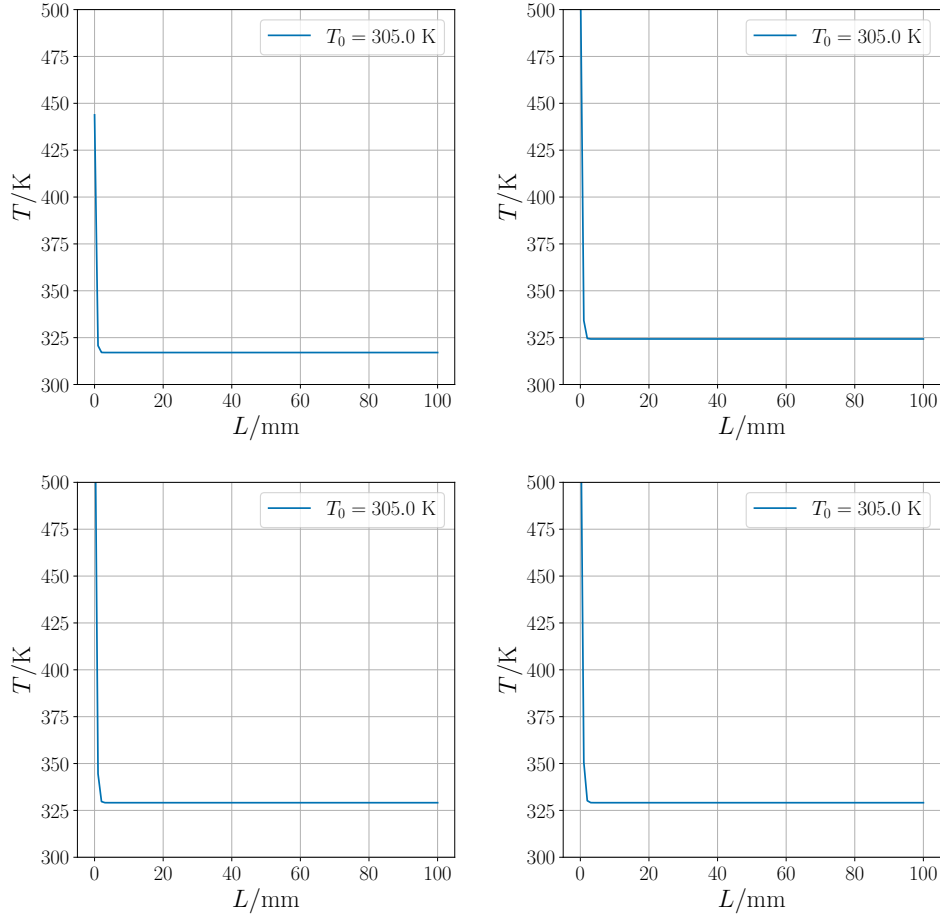


Figure 11: Temperature distributions during the heating phase. *Top row:* $t = t_h/2$, $t = 3t_h/4$. *Bottom row:* $t = t_h$, $t = 5t_h/4$.

4.4 Comparison of approximations with numerical results

We compared the results of the linear simulations with the thermostatic estimates, thereby simultaneously validating the numerical simulation; these results are summarized in Table 1. It is evident that the simulation accurately reproduces the special states of the linear

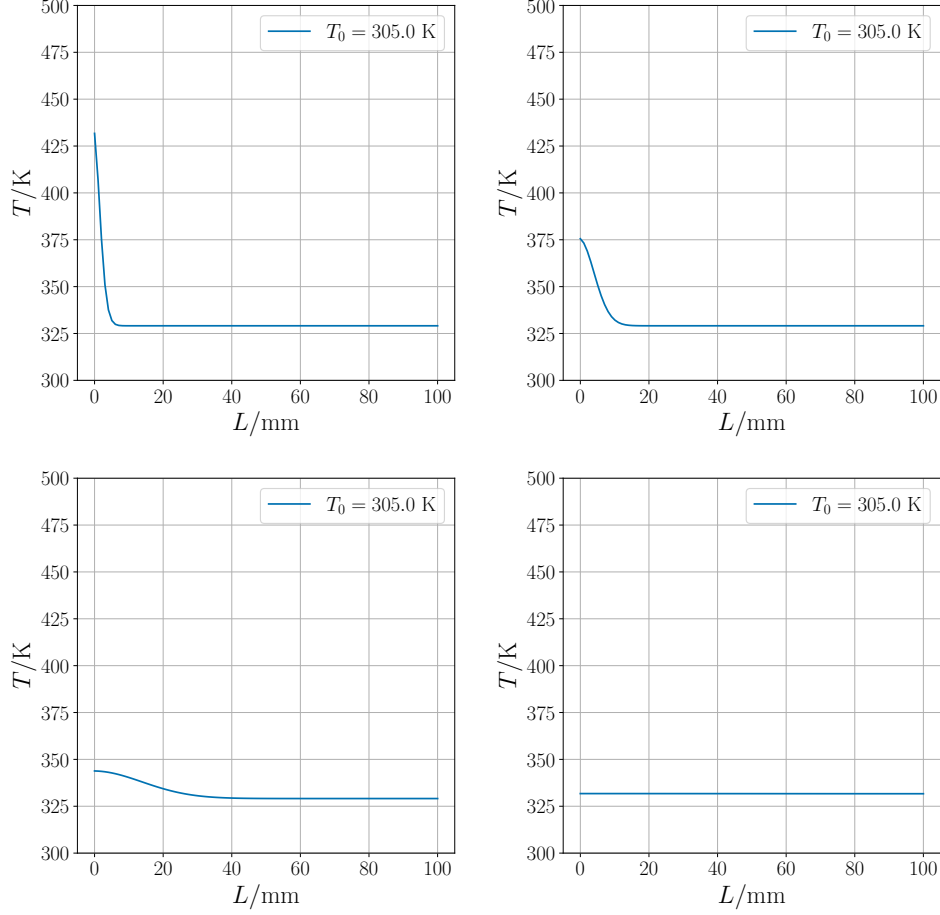


Figure 12: Temperature distributions during the diffusion phase. *Top row:* $t = \tau_{\text{diff}}/5000$, $t = \tau_{\text{diff}}/1000$. *Bottom row:* $t = \tau_{\text{diff}}/100$, $t = \tau_{\text{diff}}$.

cases, namely the state after the heating phase and the homogeneous final state. The first case was calculated using 3.85 mW heating power as in the Straub experiment, while the second case used a 3.85 W heating power. In both cases, we initiated the simulation under the initial conditions of $p^0 = 7.38$ MPa, $T^0 = 305$ K, and a heating duration of $t_h = 10$ s. In the third case, we applied the same quantity of heat input $Q = 38.5$ J as in the second case, but the heating duration was $t_h = 2$ s.

Q/J	t/s	$\frac{T^{0+}-T^0}{\text{K}}$		$\frac{T^\infty-T^0}{\text{K}}$		$\frac{p^\infty-p^0}{\text{Pa}}$	
		calculated	simulated	calculated	simulated	calculated	simulated
0.0385	10	0.0241	0.0241	0.0267	0.0267	2992.0	2991.4
38.5	10	24.1044	24.1044	26.7145	26.7142	2992041	2992040
38.5	2	24.1044	24.1044	26.7145	26.7142	2992041	2992040

Table 1: Table presenting the outcomes of various heat input scenarios

In the linear case, the process is determined solely by the heat input, assuming identical initial pressure and temperature; consequently, varying the duration of the heat input does not affect the outcome. Furthermore, the linear nature of the solved equations can be observed, as introducing a thousand times the amount of heat resulted in a thousand-fold increase in pressure and temperature, in accordance with formulas (24), (26), and (28).

4.5 Investigation of the effects of initial conditions

In this subsection, we present the influence of the initial pressure and temperature on the process. For these simulations, we selected a heat input of 38.5 J. First, we investigate five cases calculated at different initial temperatures under identical initial pressure, followed by five cases calculated at different initial pressures under identical initial temperature.

We initiated simulations at a fixed initial pressure of $p^0 = 7.38$ MPa with initial temperatures of $T^0 = 305$ K, 310 K, 315 K, 320 K, and 325 K. First, we examine the points defined by the initial conditions, specifically the values they determine for the thermophysical properties characterizing the dynamics of the process. Similarly to Figure 7, we present the locations of these initial conditions in Figures 13, 14, and 15. Examining the diagram of the term describing the temperature rise during the heating phase, it is evident that the point corresponding to the initial temperature of $T^0 = 305$ K selects a significantly lower value than the other four points. Consequently, we expect the temperature rise during the heating phase for this case to be more moderate compared to the other cases. On the diagram of the term influencing the pressure rise, it can be seen that the states defined by the initial conditions determine approximately the same value for this parameter, with the exception of the point corresponding to the initial temperature $T^0 = 305$ K. In this case, this value is approximately 10 % lower than the others; therefore, a lower pressure rise is expected. It is also noticeable that the point corresponding to the initial temperature $T^0 = 315$ K is located very close to the point where the isobar corresponding to $p = 1.00p_c$ intersects the isobar corresponding to $p = 1.20p_c$; therefore, this case may behave differently from the others. On the diagram of the term describing the temperature rise over the entire process, it can be seen that there are no significant outliers among the points determined by the initial conditions; consequently, we expect that the $T - t$ curves will arrange themselves relatively evenly above one another at the end of the diffusion process. Table 2 contains the numerical values of the thermophysical

property combinations. Ψ denotes the parameter combination describing the temperature rise during the heating phase and Θ represents the parameter combination describing the temperature rise during the diffusion phase. The table also contains the numerical values of the τ_a acoustic and τ_{diff} diffusion timescales determined by the initial conditions.

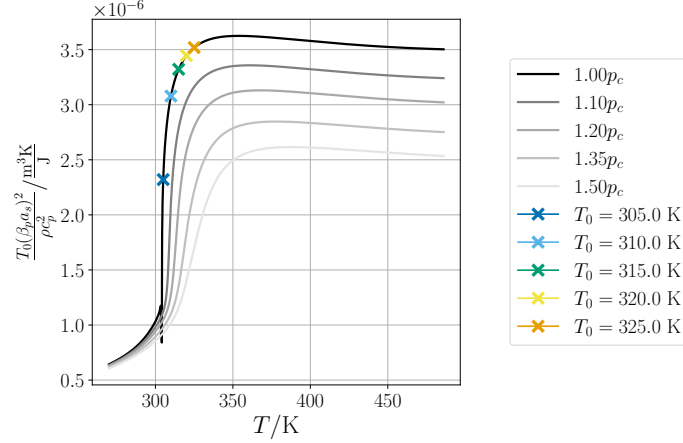


Figure 13: Thermophysical property combination describing the temperature rise during the heating phase.

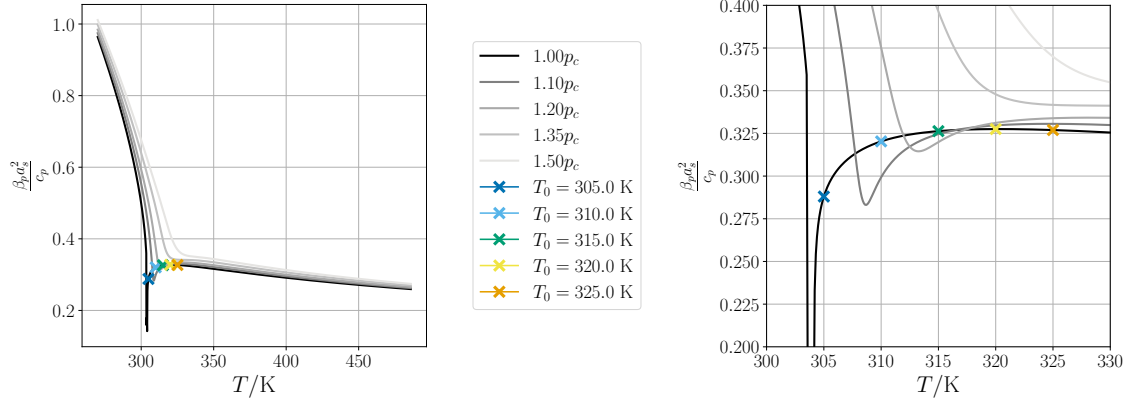


Figure 14: Thermophysical property combination describing the pressure rise.

Figures 16 and 17 illustrate the time evolution of pressure and temperature changes for the five cases. The processes developed in agreement with the expectations derived from the thermostatic estimates: the pressure rise is indeed lower for the case corresponding to an initial temperature of $T^0 = 305$ K compared to the other cases. The case corresponding to the initial temperature of $T^0 = 315$ K did not exhibit different behavior compared with the other cases, despite its proximity to an isobar corresponding to a different initial pressure. It can be observed that the temperature change during the heating phase

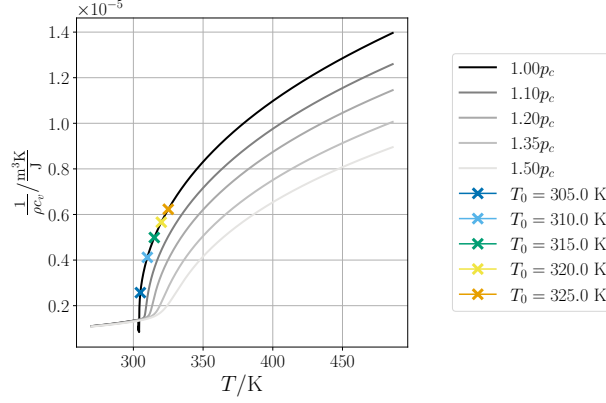


Figure 15: Thermophysical property combination describing the temperature rise during the diffusion phase.

	$\frac{\tau_a}{s}$	$\frac{\tau_{diff}}{s}$	$\frac{\Psi}{\frac{m^3 K}{J}}$	Γ	$\frac{\Theta}{\frac{m^3 K}{J}}$
$T_0 = 305 \text{ K}$	0.0005356	678208	2.32e-06	0.28801	2.57e-06
$T_0 = 310 \text{ K}$	0.0004833	265182	3.08e-06	0.32032	4.12e-06
$T_0 = 315 \text{ K}$	0.0004605	187255	3.32e-06	0.32633	4.99e-06
$T_0 = 320 \text{ K}$	0.0004445	149959	3.45e-06	0.32753	5.66e-06
$T_0 = 325 \text{ K}$	0.0004318	127194	3.52e-06	0.32696	6.23e-06

Table 2: Numerical values of the thermophysical property combinations and of the acoustic and diffusion time scales.

followed the description above. On the full time scale, the temperature change-time curves show greater separation from the case corresponding to $T^0 = 305 \text{ K}$ than we would have expected based on Figure 15. Each simulation was run for the duration of the diffusion time scale corresponding to its specific initial condition, which explains the differing plotting intervals for the individual cases.

We investigated the dependence of the process on the initial pressure at a fixed initial temperature of $T^0 = 305 \text{ K}$, using initial pressures of $p_0 = 7.38 \text{ MPa}$, 7.48 MPa , 7.58 MPa , 7.68 MPa , and 7.78 MPa . Similarly to the investigation of the initial temperature dependence, we illustrate the thermophysical property combinations corresponding to these initial conditions in Figures 18, 19, and 20. Examining the thermophysical property combination describing the temperature rise during the heating phase, it is evident that the value corresponding to the initial pressure $p^0 = 7.38 \text{ MPa}$ is slightly larger than the others; thus, we expect the temperature rise during the heating

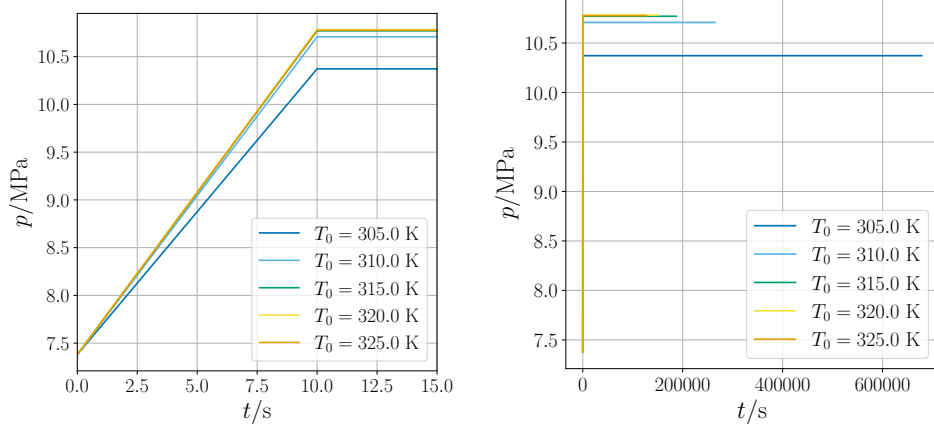


Figure 16: Pressure as a function of time for different initial temperatures.

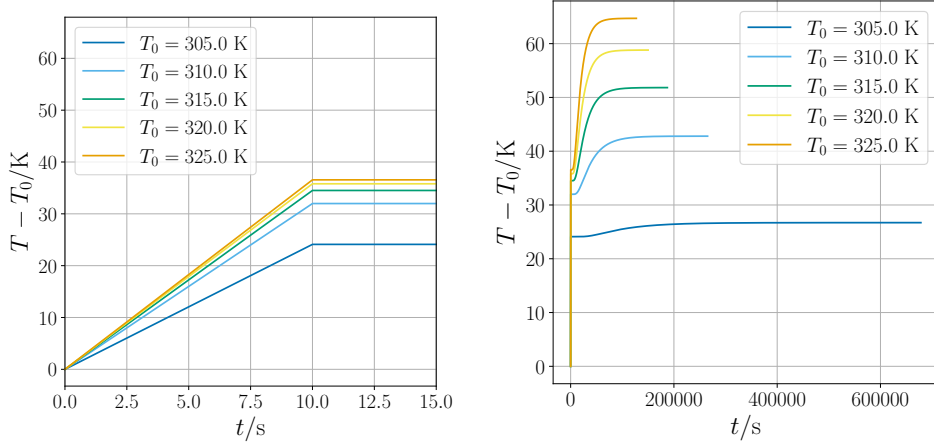


Figure 17: Temperature change as a function of time for different initial temperatures.

phase to be greater in this case compared to the others. Furthermore, based on the diagram, it is expected that the temperature rise during the heating phase will be nearly identical for the cases with initial pressures of $p^0 = 1.03p_c$, $p^0 = 1.04p_c$, $p^0 = 1.05p_c$. Based on the diagram of the term influencing the pressure rise, we expect the pressure rise to be nearly identical for the cases with initial pressures of $p^0 = 1.00p_c$, $p^0 = 1.03p_c$. Regarding the term influencing the temperature rise over the entire process, we can expect that in the cases corresponding to initial pressures $p^0 = 1.03p_c$, $p^0 = 1.04p_c$, $p^0 = 1.05p_c$ the temperature rise over the entire process will be nearly identical. Analogous to the constant initial pressure cases, Table 3 contains the numerical values of the thermophysical property combinations and of the acoustic and diffusion time scales for the different initial pressures.

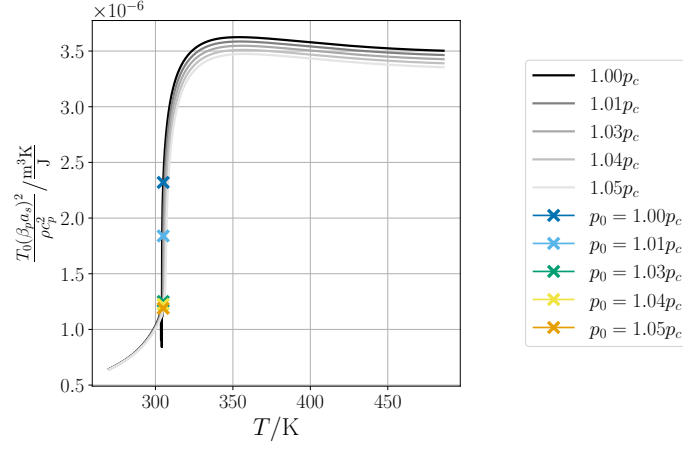


Figure 18: Thermophysical property combination describing the temperature rise during the heating phase.

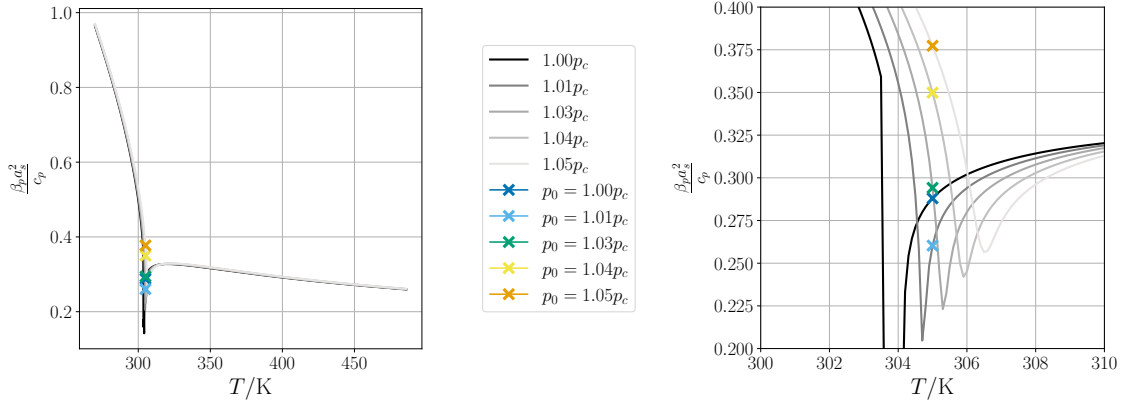


Figure 19: Thermophysical property combination describing the pressure rise.

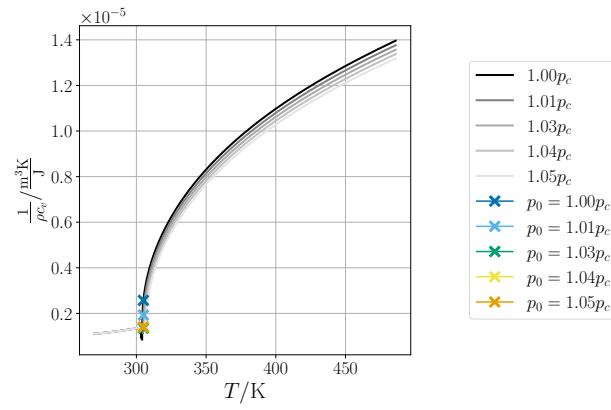


Figure 20: Thermophysical property combination describing the temperature rise during the diffusion phase.

	$\frac{\tau_a}{s}$	$\frac{\tau_d}{s}$	$\frac{\Psi}{\frac{m^3 K}{J}}$	Γ	$\frac{\Theta}{\frac{m^3 K}{J}}$
$p_0 = 7.38 \text{ MPa}$	0.0005356	678208	2.32e-06	0.28801	2.57e-06
$p_0 = 7.48 \text{ MPa}$	0.0005694	1255735	1.84e-06	0.26032	1.93e-06
$p_0 = 7.58 \text{ MPa}$	0.0005203	1142664	1.25e-06	0.29414	1.35e-06
$p_0 = 7.68 \text{ MPa}$	0.0004480	693347	1.22e-06	0.34989	1.40e-06
$p_0 = 7.78 \text{ MPa}$	0.0004165	577692	1.19e-06	0.37733	1.41e-06

Table 3: Numerical values of the thermophysical property combinations and of the acoustic and diffusion time scales.

Figures 21 and 22 present the results of the five cases. The processes unfolded as expected: the pressure rise was indeed nearly identical in the cases of $p^0 = 7.38 \text{ MPa}$ and $p^0 = 7.58 \text{ MPa}$, while the temperature change during the heating phase was nearly identical for the cases with initial pressures of $p^0 = 7.58 \text{ MPa}$, $p^0 = 7.68 \text{ MPa}$ and $p^0 = 7.78 \text{ MPa}$. Contrary to our expectations, during the diffusion phase, the curve corresponding to the initial condition $p^0 = 7.58 \text{ MPa}$ crosses the curves corresponding to initial pressures of $p^0 = 7.68 \text{ MPa}$ and $p^0 = 7.78 \text{ MPa}$. The reason for this is that while a smaller temperature change occurs during the diffusion phase for the initial pressure of $p^0 = 7.58 \text{ MPa}$, a larger temperature change occurred at the beginning of the process for this initial pressure.

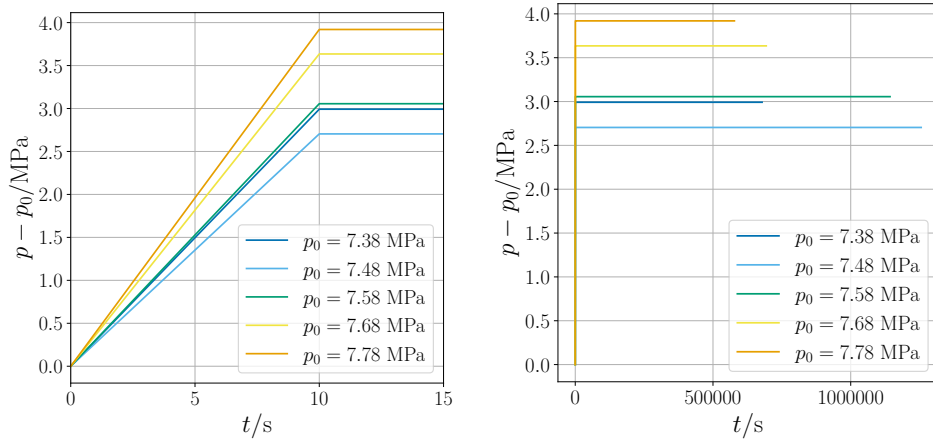


Figure 21: Pressure change as a function of time for different initial pressures.

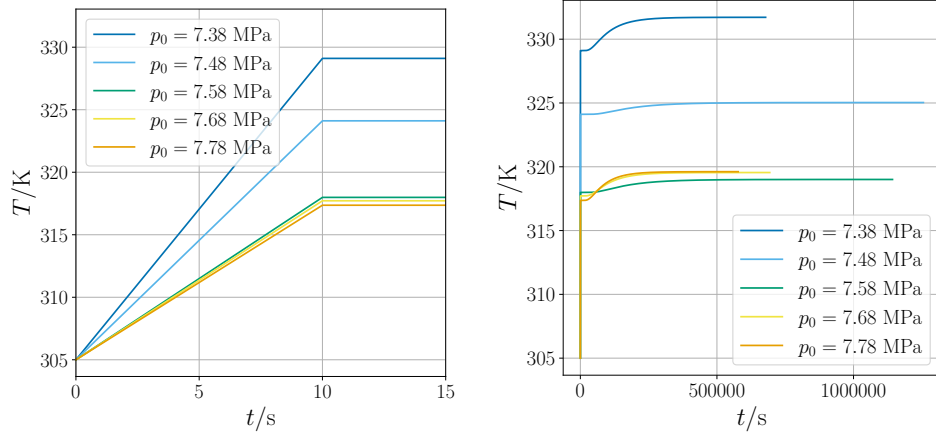


Figure 22: Temperature as a function of time for different initial pressures.

5 Nonlinear case

The nonlinearity of equations (7)–(9) is caused not only by the $T\beta_p \frac{dp}{dt}$ term appearing in the heat conduction equation but also by the nonlinearities arising from material behavior, which we presented earlier. To accurately simulate these material nonlinearities, we employed state-dependent thermophysical properties in the simulations, determining this state dependence using the previously introduced Span–Wagner equation of state.

The Span–Wagner equation of state expresses pressure as a function of density; however, during the numerical simulation, pressures and temperatures are calculated at every step. Therefore, we must determine the density corresponding to a given pressure using a root-finding algorithm based on the $p(T, \rho)$ function at a known temperature. Initially, we used the bisection method to search for roots, but this proved to be too slow. Later, we employed the root-finding algorithm based on Newton’s method from the Python SciPy library. If the derivative of the target function is not known, the built-in subroutine applies the secant method. Both methods possess more favorable convergence properties than the bisection method. Using the derivatives of the dimensionless Helmholtz free energy provided in the article [8], we derived the function $\left. \frac{\partial p}{\partial \rho} \right|_T$. Applying this function within Newton’s method increases the order of convergence to 2. We compared this algorithm with the algorithm based on the secant method. We observed that while the algorithm based on Newton’s method could achieve greater accuracy, it had a slower execution time than the secant method-based algorithm. Since the algorithm can produce results at most as accurate as the Span–Wagner equation of state it solves, we prioritized execution speed and proceeded to work with the secant method-based algorithm in the subsequent steps.

5.1 Discretization

To simulate the nonlinear cases, we discretized the nonlinear equations (7)–(9) directly:

$$T_{n+\frac{1}{2}}^{j+1} = T_{n+\frac{1}{2}}^j + \frac{1}{\rho_{n+\frac{1}{2}}^j c_{p,n+\frac{1}{2}}^j} \frac{\Delta t}{\Delta x} (\dot{q}_n^j - \dot{q}_{n+1}^j) + \frac{T_{n+\frac{1}{2}}^j \beta_{p,n+\frac{1}{2}}^j}{\rho_{n+\frac{1}{2}}^j c_{p,n+\frac{1}{2}}^j} (p^{j+1} - p^j), \quad (34)$$

$$p^{j+1} = p^j + \frac{\sum_{n=0}^{N-1} \rho_{n+\frac{1}{2}}^j \beta_{p,n+\frac{1}{2}}^j (T_{n+\frac{1}{2}}^{j+1} - T_{n+\frac{1}{2}}^j) \frac{\Delta x}{\Delta t}}{\sum_{n=0}^{N-1} \rho_{n+\frac{1}{2}}^j \chi_{T,n+\frac{1}{2}}^j \Delta x} \Delta t, \quad (35)$$

$$\dot{q}_n^{j+1} = \lambda_n^j \frac{T_{n-\frac{1}{2}}^{j+1} - T_{n+\frac{1}{2}}^{j+1}}{\Delta x}, \quad (36)$$

where the material properties must be updated cell-by-cell at every time step; for example,

$$\beta_{p,n+\frac{1}{2}}^j = \beta_p \left(T_{n+\frac{1}{2}}^{j+1}, p^{j+1} \right). \quad (37)$$

The calculation procedure is similar to that of the linear case. We first determine the thermophysical properties at a fixed j time step. Then, by iteratively solving equations (34) and (35), we obtain the values $T_{n+\frac{1}{2}}^{j+1}$ and p^{j+1} . Finally, we determine the heat flux density, and the calculation cycle repeats.

5.2 Numerical investigation

In the following non-linear simulations, we employed 20 cells along the x axis.

We examined the process again under the initial conditions of $p^0 = 7.38$ MPa and $T^0 = 305$ K, with a heating duration of $t_h = 10$ s and a heating power of $P = 3.85$ mW as in the Straub experiment. Figures 23 and 23 compare the linear and non-linear cases. It is evident that under the low-energy excitation applied in the experiment, nonlinear effects do not manifest; the curves of the nonlinear case overlap perfectly with those of the linear case. Table 4 presents the temperature and pressure values after the heating phase, as well as the equilibrium temperature, for both linear and nonlinear calculations, also indicating the results of the analytical estimates. Although not significant, a deviation in the pressure rise is observable even at such low excitation.

Next, we performed the simulation using the same initial conditions but with a thousand-fold increase in heating power. We expect that the higher excitation will induce

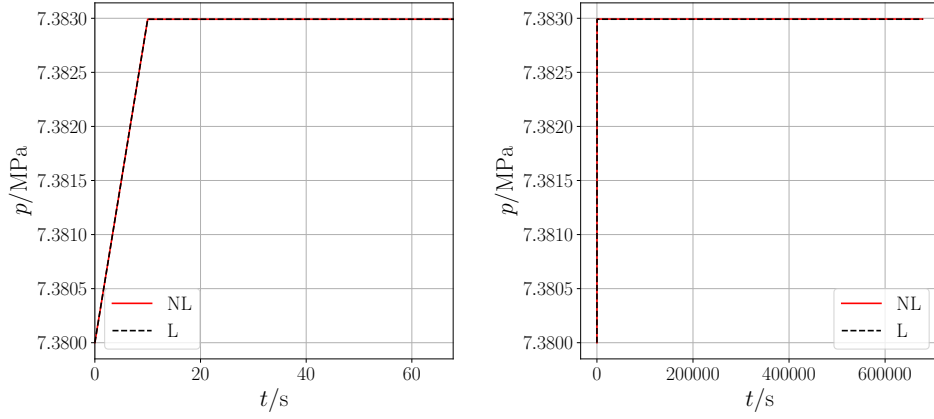


Figure 23: Pressure as a function of time in the nonlinear (NL) and linear (L) cases.

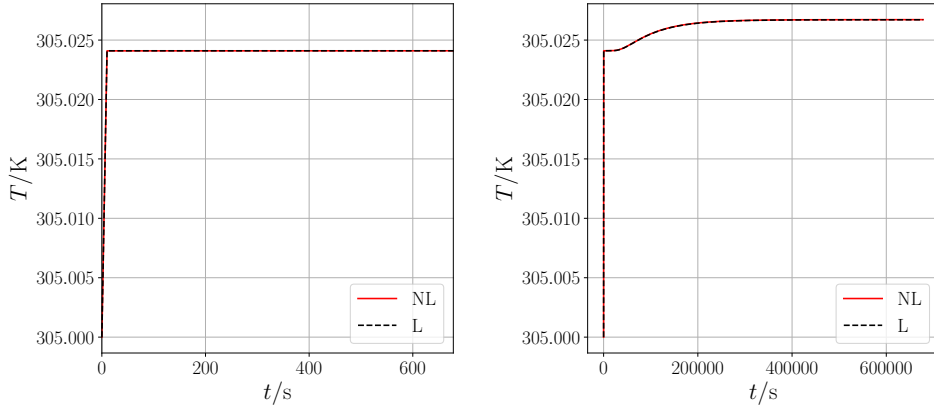


Figure 24: Temperature as a function of time in the nonlinear (NL) and linear (L) cases.

nonlinear material behavior, resulting in significant deviations between the nonlinear and linear cases in the diagrams. As before, we plotted the temperature calculated in the cell furthest from the heat input as a function of time. The results are illustrated in Figures 25 and 26 and summarized in Table 5. In agreement with our expectations, the increased excitation was sufficient to induce material nonlinearities. Observe that both the pressure and temperature changes during the heating phase fall short of the linear predictions; however, in the homogeneous final state, both quantities exceed them. It is also worth noting that the pressure change continues even after the conclusion of the heating phase, in contrast to the linear case. Figure 27 illustrates the trajectories of the linear and nonlinear processes in the p - T thermodynamic state space. Note that in the linear case, during the heat input (while the pressure is changing), the process path on the p - T diagram is a straight line, whereas in the nonlinear process, it deviates from this. The explanation for these observations lies in the variation of thermophysical properties

	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{p^\infty-p^0}{\text{Pa}}$	$\frac{p^\infty-p^0}{\text{Pa}}$
	calculated	simulated	calculated	simulated	calculated	simulated
Linear case	0.0241	0.0241	0.0267	0.0267	2992.0	2991.4
Nonlinear case	0.0241	0.0241	0.0267	0.0267	2992.0	2992.9

Table 4: Comparison of results from linear and nonlinear simulations.

throughout the process.

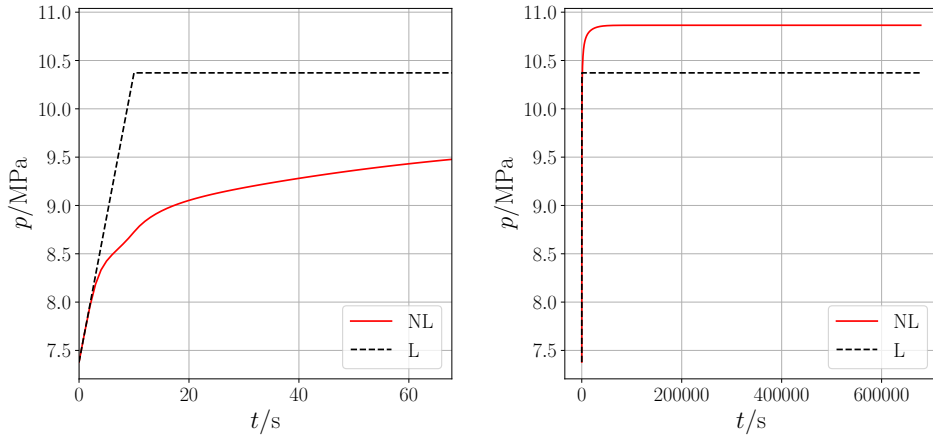


Figure 25: Pressure as a function of time in the nonlinear (NL) and linear (L) cases.

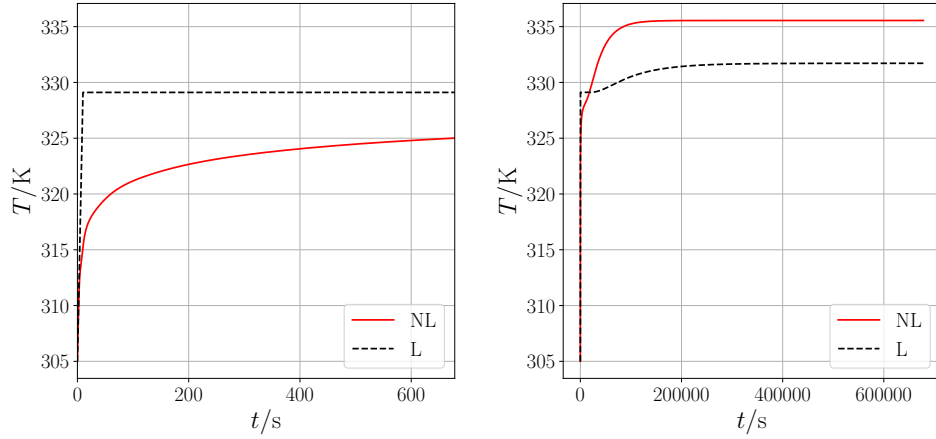


Figure 26: Temperature as a function of time in the nonlinear (NL) and linear (L) cases.

Figures 28, 29, and 30 illustrate the evolution of the dominant terms influencing the dynamics along the process path. Similar to the previous instances, we plotted the evolution of these terms using thermophysical properties evaluated in the cell furthest from the

	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^{0+}-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{T^\infty-T^0}{\text{K}}$	$\frac{p^\infty-p^0}{\text{Pa}}$	$\frac{p^\infty-p^0}{\text{Pa}}$
	calculated	simulated	calculated	simulated	calculated	simulated
Linear case	24.10	24.10	26.71	26.71	2992040.7	2992040.1
Nonlinear case	24.10	10.23	26.71	30.55	2992040.7	3484634.4

Table 5: Comparison of results from linear and nonlinear simulations.

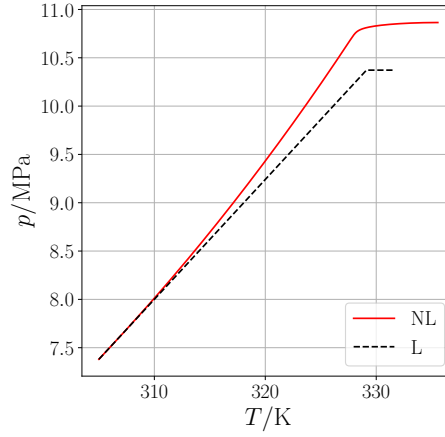


Figure 27: Trajectories of the linear (L) and nonlinear (N) processes in the p - T thermodynamic state space.

location of the heat input. The value of the term describing the temperature rise during the heat input decreases during the heating phase and subsequently converges to the isobar corresponding to the equilibrium pressure. This decrease observed at the beginning of the process explains why the temperature rise during the heating phase is more significant in the linear case than in the nonlinear case. The value of the thermophysical property combination describing the pressure rise increases throughout the heating phase and subsequently converges to the isobar corresponding to the equilibrium pressure. Based on this, one would infer that the pressure rise during the process would be more significant in the nonlinear case than in the linear case. In contrast, the numerical investigations show a deceleration of the pressure rise, followed by an overshoot compared to the linear case. This issue requires further investigation. The value of the term describing the temperature rise during the diffusion phase does not change significantly during the heating phase; then, similarly to the other two parameters, it converges to the isobar corresponding to the equilibrium pressure, along which its value then increases significantly until the end of

the process. This explains the increase in final temperature compared to the linear case.

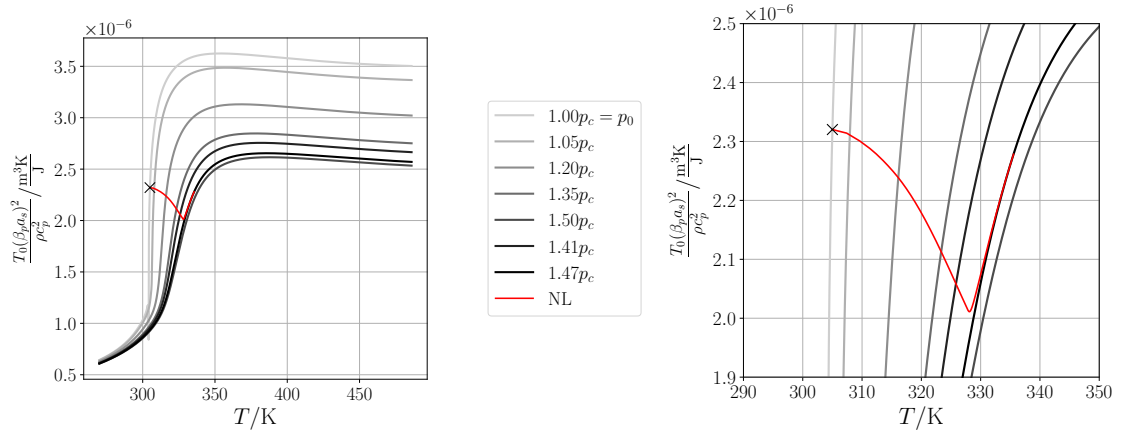


Figure 28: Trajectory of the thermophysical property combination describing the temperature rise during the heating phase.

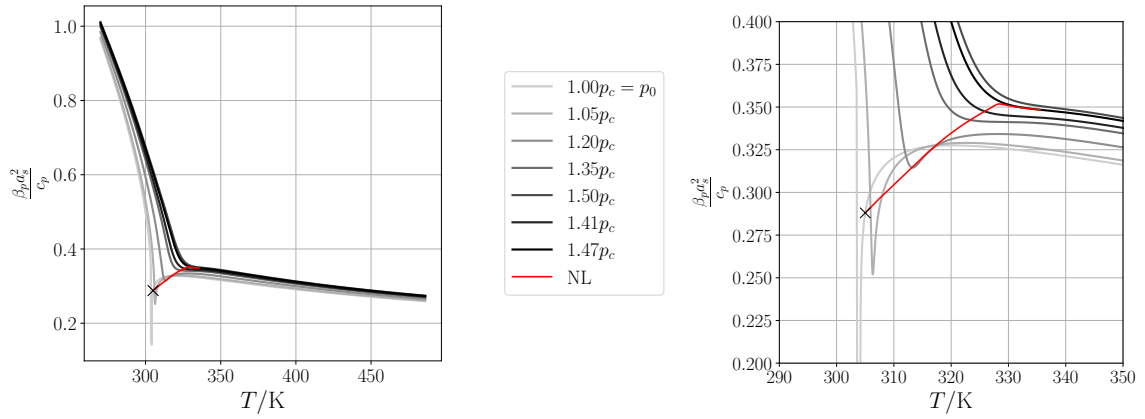


Figure 29: Trajectory of the thermophysical property combination describing the pressure rise.

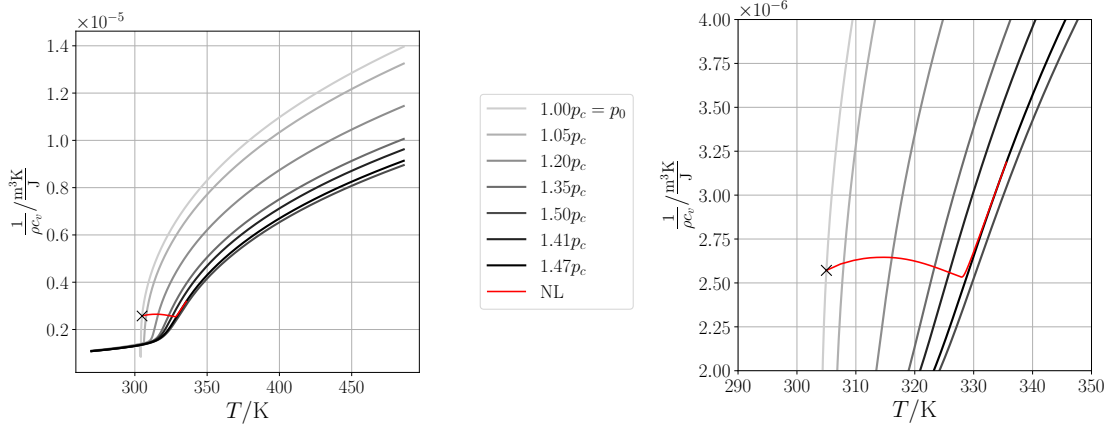


Figure 30: Trajectory of the thermophysical property combination describing the temperature rise during the diffusion phase.

Figures 31 and 32 illustrate the time evolution of the temperature distributions in the linear and nonlinear cases. The diagrams show that during the heating phase, the entire fluid heated up at a significantly higher rate in the linear case than in the nonlinear case; that is, the piston effect is more dominant in the linear case. In contrast, the diffusion process is faster in the nonlinear case; this behavior was already observable from Figure 26. The explanation for this observation lies in the evolution of the thermal conductivity and thermal diffusivity along the process, which is illustrated in Figures 33 and 34. Although the value of thermal conductivity decreases along the process, the value of thermal diffusivity continuously increases. Even though the process is nonlinear, the dynamics of the diffusion phase are governed by the thermal diffusivity; thus, its continuously increasing value results in increasingly faster heat conduction.

5.3 Influence of heating duration

In accordance with our expectations, in the linear case only the amount of heat input affected the outcomes under identical initial conditions; that is, varying the duration of the heating phase had no influence on the simulation results. We will now investigate whether altering the duration of the heating phase induces any changes in the nonlinear case. Keeping the total heat input constant, we selected heating durations of 10 s, 2 s and 100 s. The applied heat flux density at the boundary was adjusted accordingly. Remarkably, different heating durations did not result in significant differences in the trajectories on the p - T diagram (see Figures 35 and 36), even in these nonlinear cases.

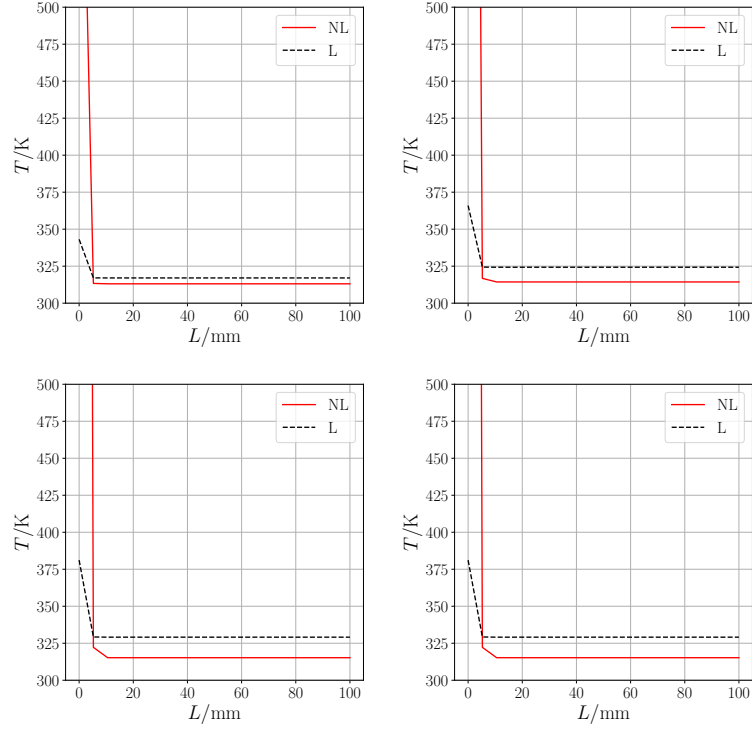


Figure 31: Temperature distributions during the heating phase. *Top row:* $t = t_h/2$, $t = 3t_h/4$. *Bottom row:* $t = t_h$, $t = 5t_h/4$.

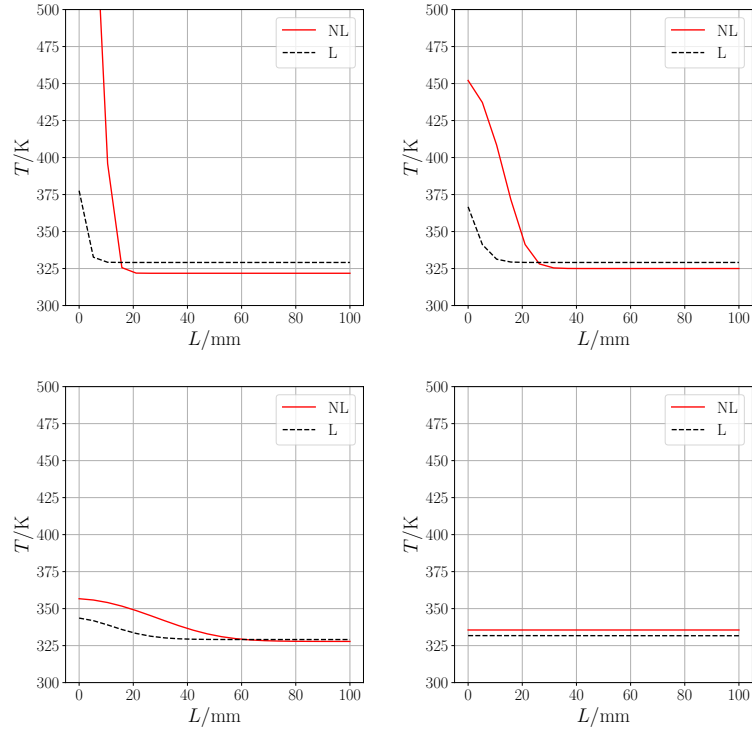


Figure 32: Temperature distributions during the diffusion phase. *Top row:* $t = \tau_{\text{diff}}/5000$, $t = \tau_{\text{diff}}/1000$. *Bottom row:* $t = \tau_{\text{diff}}/100$, $t = \tau_{\text{diff}}$.

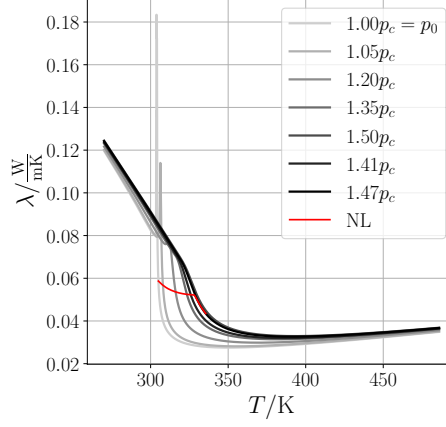


Figure 33: Trajectory of the thermal conductivity throughout the process.

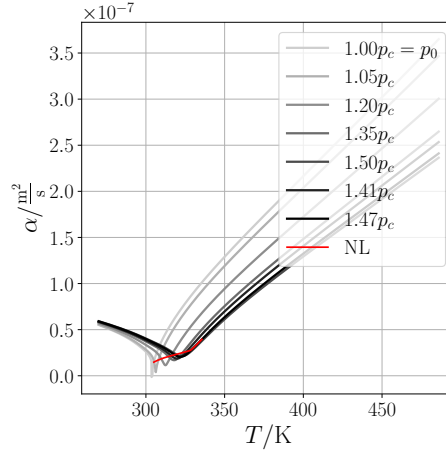


Figure 34: Trajectory of the thermal diffusivity throughout the process.

The reason for this may be that the differences in heating duration are negligible relative to the acoustic and diffusion time scales. To test a more extreme scenario, an additional simulation with a heating duration of 1000 s was performed. This duration is only two orders of magnitude shorter than the diffusion timescale; therefore, if the heating duration influences the nonlinear behavior, a more pronounced deviation in the trajectory should be observable compared with the previous cases. Figure 37 shows the trajectory of this case on the p - T diagram together with the trajectory of the 10 s heating duration case. It can be observed that even this case, the trajectory corresponding to the 1000 s heating duration differs only marginally from that of the 10 s heating duration case during the heating phase and separates from it only during the diffusion phase. This separation arises because, in the 1000 s heating duration case the pressure and temperature increases are more pronounced, leading to a deviation of the trajectory during the diffusion phase.

In the future, we plan to investigate cases with extremely short heating durations as well; for example, cases in which the heating duration is only 1–2 orders of magnitude longer than the acoustic time scale.

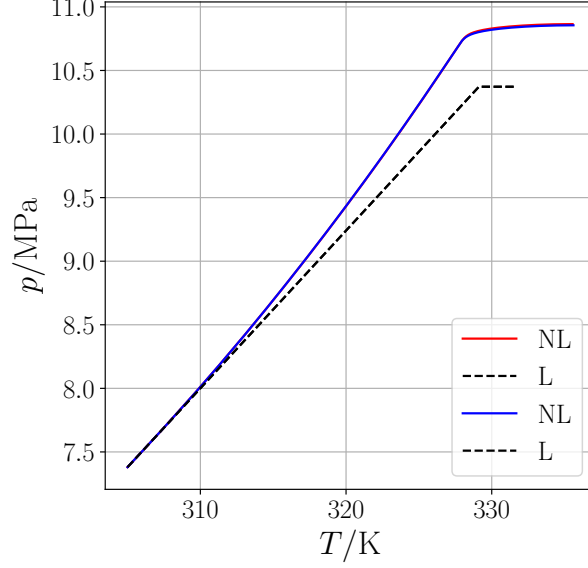


Figure 35: Trajectories of the linear (L) and nonlinear (NL) processes in the p – T thermodynamic state space. The case with a heating duration of $t_h = 10$ s is shown in red, while the case with a heating duration of $t_h = 2$ s is shown in blue.

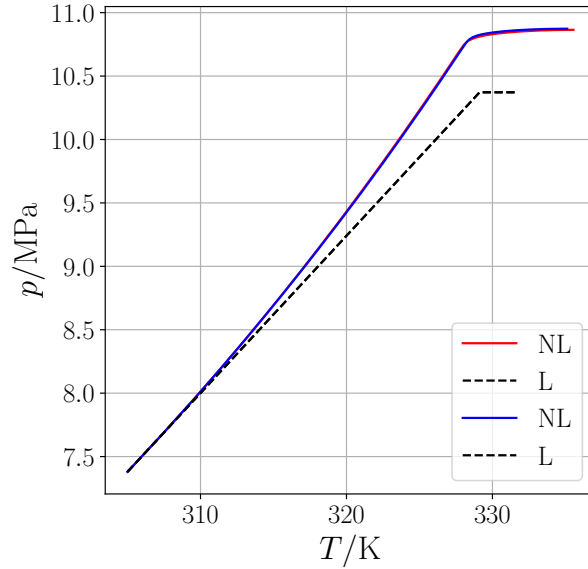


Figure 36: Trajectories of the linear (L) and nonlinear (NL) processes in the p – T thermodynamic state space. The case with a heating duration of $t_h = 10$ s is shown in red, while the case with a heating duration of $t_h = 100$ s is shown in blue.

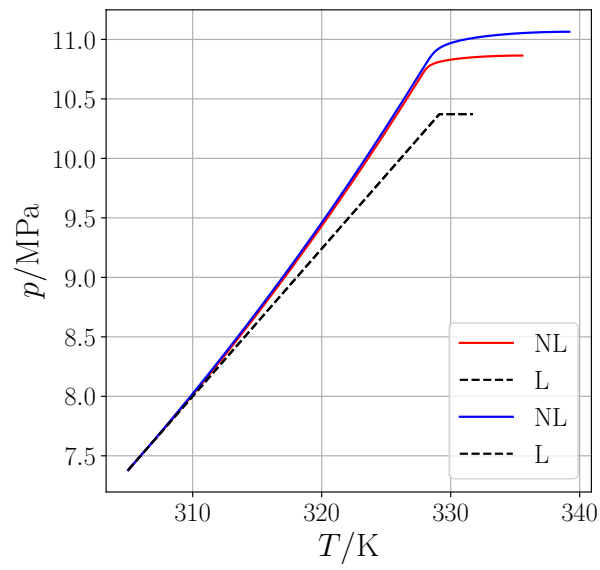


Figure 37: Trajectories of the linear (L) and nonlinear (NL) processes in the p - T thermodynamic state space. The case with a heating duration of $t_h = 10$ s is shown in red, while the case with a heating duration of $t_h = 1000$ s is shown in blue.

6 Summary and outlook

At the beginning of the thesis, we presented the material behavior in the vicinity of the critical point through the example of carbon dioxide, as well as the phenomenon of the piston effect resulting from the significant compressibility of fluids near the critical point. We presented the governing equations of the phenomenon using the homogeneous pressure approximation by Boukari. We performed the linearization and discretization of this model, and subsequently presented the phenomenon in detail through the numerical simulation of a representative case. We explored the dependence of the linear case on initial conditions by examining the dominant terms inherent in the linearized equations, and demonstrated their effects through the results of multiple numerical simulations.

We also performed the discretization of the nonlinear equations and compared the nonlinear simulation with the linear one through a representative case. We found that inducing nonlinear material behavior requires significantly higher excitation than that applied during the experimental demonstration of the phenomenon conducted within the framework of the Spacelab D2 mission; otherwise, the nonlinear case yields results nearly identical to the linear case. Thereby, we also confirmed that the assumption of linearity during the evaluation of the experiment was valid. We also examined the dependence of the nonlinear equations on the heating duration and found that this dependence was not significant for the cases investigated, particularly during the heating phase dominated by the piston effect. A notable dependence was observed only during the diffusion phase. Examining more extreme cases is necessary to confidently conclude that the nonlinear equations do not exhibit a significant dependence on the heating duration.

Further investigations are required to map the dependence of the nonlinear equations on initial conditions. In the future, we plan to identify initial conditions for which both progressive and degressive nonlinearities become apparent during specific phases of the process—namely, during the piston effect dominated heating phase and during the diffusion phase. This requires further analysis of the parameters (24)–(28), for which Figure 7 offers additional guidance.

Finally, Figure 38 depicts the nonlinear process we investigated on a p – T diagram, on which we have also indicated the Widom lines. It can be observed that the nonlinear process approaches the curve corresponding to the minimum of the isentropic speed of

sound. In the future, it would be worthwhile to initiate nonlinear simulations with initial conditions such that the processes intersect certain Widom lines. These processes may shed light on further interesting phenomena. Similarly to the Widom lines, we can plot the extrema and inflection points of the dominant thermophysical property combinations (see Figure 39), which may shed further light on how initial conditions can be chosen such that the resulting process exhibits both progressive and degressive characteristics.

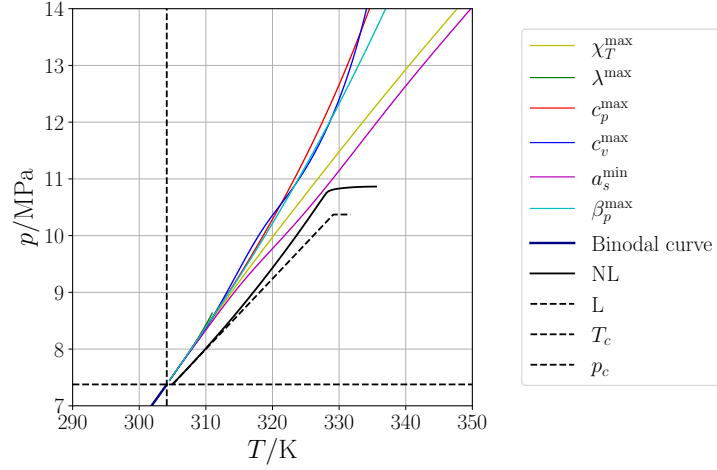


Figure 38: The investigated nonlinear process and the Widom lines on the p - T diagram.

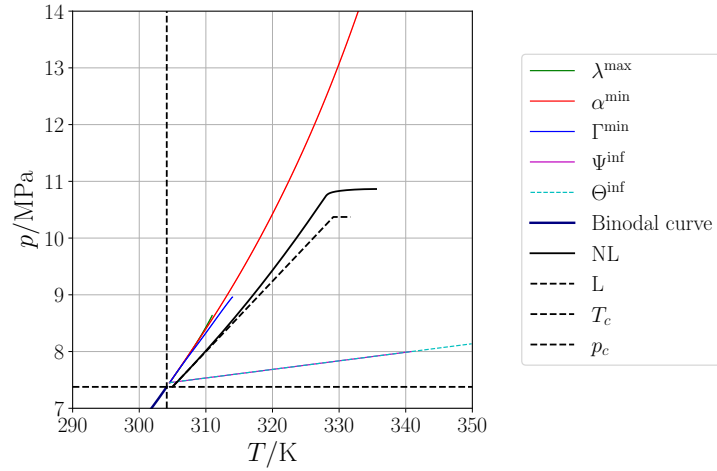


Figure 39: The investigated nonlinear process together with the extrema and inflection points of the dominant thermophysical property combinations on the p - T diagram. The superscript ^{inf} denotes the inflection points.

7 Összefoglalás

A dolgozat elején szén-dioxid példáján keresztül bemutattuk az anyagi viselkedést a kritikus pont környezetében, valamint a kritikus pont környezetében a fluidumok jelentős összenyomhatóságából adódó dugattyúhatás jelenségét. Bemutattuk a jelenség leíró egyenleteit a Boukari-féle homogén nyomás közelítéssel. Elvégeztük ezen modell linearizálását és diszkretizációját, majd pedig részletesen bemutattuk a jelenséget egy jellemző eset numerikus szimulációja által. Felderítettük a lineáris eset kezdeti feltételektől való függését a linearizált egyenletekben rejlő domináns tagok vizsgálata által, és több numerikus szimuláció eredményein keresztül bemutattuk hatásukat.

A nemlineáris egyenletek diszkretizációját is elvégeztük, és egy jellemző eseten keresztül összehasonlítottuk a nemlineáris szimulációt a lineárisal. Kivizsgáltuk, hogy a nemlinearitások megébredéséhez jelentősebb gerjesztés szükséges, mint amit a jelenség kísérleti kimutatásánál alkalmaztak a Spacelab D2 misszió keretein belül, ellenkező esetben a nemlineáris eset a lineáris esettel szinte azonos eredményt ad. Ezzel egyben azt is igazoltuk, hogy a kísérlet kiértékelésénél a linearitás feltételezése helytálló volt. Megvizsgáltuk a nemlineáris egyenletek hőbeviteli időtől való függését is, és megállapítottuk, hogy az általunk vizsgált esetekben nem volt számottevő a hőbeviteli időtől való függés a hőbevitel során, ahol a dugattyúhatás dominál. Számottevő hőbeviteli időtől való függés csak a diffúziós szakaszon volt megfigyelhető.

A nemlineáris egyenletek kezdeti feltételtől való függésének feltérképezéséhez további vizsgálatok szükségesek. A későbbiekben tervezzük olyan kezdeti feltételek megtalálását, hogy a folyamat egyes szakaszain, azaz a dugattyúhatás érvényesülésekor, illetve a diffúziós szakaszon látszódnak egyaránt progresszív és degresszív nemlinearitások. Ehhez a (24)–(28) paraméterek további elemzése szükséges, amihez a 7. ábra nyújt további segítséget.

Végezetül a 38. ábra az általunk vizsgált nemlineáris folyamatot ábrázolja p – T diagramon, amin a Widom-vonalakat is feltüntettük. Megfigyelhető, hogy a nemlineáris folyamat közel kerül a hangsebesség minimum görbéjéhez. A jövőben érdemes lenne olyan kezdeti feltételekkel indítani a nemlineáris szimulációkat, hogy a folyamatok metsszenek el bizonyos Widom-vonalakat. Ezek a folyamatok további érdekes jelenségekre világíthatnak rá.

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